

**TRIBHUVAN UNIVERSITY**  
**INSTITUTE OF ENGINEERING**  
**PULCHOWK CAMPUS**



**PROJECT REPORT**

*on*

**Dual-Setting Directional Recloser and Fuse Coordination  
in Unbalanced Distribution Networks  
With Distributed Generation**

**Submitted by**

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# ABSTRACT

Distributed generation (DG) changes the magnitude and the direction of the fault current in medium-voltage distribution feeders and can therefore invalidate a conventional, fixed-setting recloser–fuse coordination. This project independently implements and evaluates the dual-setting directional recloser (DSDR) method of Yousaf *et al.* on the IEEE 13-node test feeder. The earlier phase of the project used MATLAB/Simulink; the complete study has now been rebuilt in **DIgSILENT PowerFactory 2021**, driven by Python scripts, so that every table and figure is regenerated from the model by one command.

The model uses the feeder of the DIgSILENT example library with the 115/4.16 kV substation transformer, a 4.05 MVA synchronous DG at node 692 with its full machine data, the manufacturer curves of the two reclosers (GE IAC77B801A and GE/Alstom CDG34) and library fuse curves (Gould-Shawmut A055C). Unbalanced load flow, short-circuit calculation for all fault types and phase combinations, and an electromagnetic-transient (EMT) simulation of the reclosing sequence were carried out.

The fault levels of the model agree with the IEEE 13-node short-circuit benchmark within about 2 %. Without DG the scheme is coordinated in all 39 node/fault-type cells. With the DG and a conventional single-setting recloser R2, coordination is held in only 24 of 39 cells. With R2 as a DSDR and the fuse revision of the method, coordination is held in 38 of 39 cells. With the same fuses and a single-setting R2 only 31 cells are held, so the dual setting itself restores seven cells. One cell (line-to-ground fault at 692) cannot be restored with the available dial range, and faults downstream of R2 remain fed directly by the DG. PowerFactory’s own relay and fuse models confirm the calculated operating times within 1.22 % over 1896 operating times.

The report documents the model, the settings, the results and the work that remains: the extension to the IEEE 34-node feeder. The comparison with the reference paper is given separately in Appendix E.

**Keywords:** distributed generation, dual-setting directional recloser, fuse saving, recloser–fuse coordination, IEEE 13-node feeder, DIgSILENT PowerFactory.

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## LIST OF ABBREVIATIONS

| Abbreviation | Meaning                                 |
|--------------|-----------------------------------------|
| CT           | Current Transformer                     |
| CTI          | Coordination Time Interval              |
| DG           | Distributed Generation                  |
| DSDR         | Dual-Setting Directional Recloser       |
| EMT          | Electromagnetic Transient (simulation)  |
| LG           | Line-to-Ground fault                    |
| LL           | Line-to-Line fault                      |
| LLG          | Line-to-Line-to-Ground fault            |
| LLL          | Three-Phase fault                       |
| MMT          | Minimum Melting Time                    |
| OLF          | Overload Factor                         |
| PD           | Protection Device                       |
| PF           | DIgSILENT PowerFactory                  |
| R1           | Feeder-Head Recloser                    |
| R2           | Mid-Line Recloser (the DSDR)            |
| RMS          | Root Mean Square                        |
| SBDG         | Synchronous-Based Distributed Generator |
| TCC          | Time-Current Characteristic             |
| TCT          | Total Clearing Time                     |
| TDS          | Time Dial Setting                       |
| TMS          | Time Multiplier Setting                 |

## LIST OF SYMBOLS

| Symbol             | Description                                                              |
|--------------------|--------------------------------------------------------------------------|
| $I_f$              | Fault current seen by a protection device                                |
| $I_p$              | Pickup current of a recloser                                             |
| $I_s$              | Plug (current) setting of the CDG relay                                  |
| $I_{nom}$          | Rated (load) current                                                     |
| $M$                | Multiple of pickup, $I_f/I_p$                                            |
| $TDS, TMS$         | Time-dial / time-multiplier setting                                      |
| $t_F, t_D$         | Fast and delayed operating time of a recloser                            |
| $t_{MMT}, t_{TCT}$ | Minimum-melting and total-clearing time of a fuse                        |
| $A, B, n$          | Constants of the inverse-time curve                                      |
| $a_i, b_i$         | Coefficients of the fuse characteristic                                  |
| $k, l, m$          | Index of the recloser, of a downstream and of an upstream fault location |
| $z, i$             | Number of series fuses in the fault path; position of a fuse in it       |
| $Z_f$              | Fault impedance                                                          |
| $X_d, X'_d, X''_d$ | Direct-axis synchronous, transient and subtransient reactance            |
| $X_q, X'_q, X''_q$ | Quadrature-axis reactances                                               |
| $R_a$              | Armature resistance                                                      |

# CHAPTER ONE: INTRODUCTION

## 1.1 Background

Overcurrent protection of radial distribution networks is traditionally coordinated on the assumption that the fault current flows from the substation towards the loads. A recloser upstream of a fuse first operates on a fast characteristic, to clear a temporary fault before the fuse melts, and then on a delayed characteristic, to let the fuse clear a permanent fault. This *fuse-saving* arrangement only works while the fuse curve lies between the two recloser curves over the whole range of fault currents.

Distributed generation changes this. A DG downstream of a recloser feeds current towards a fault upstream of it, so the recloser carries current in the reverse direction, and a fuse between the two sources carries the sum of both contributions while each recloser carries only one. The current seen by a protection device then depends on the fault location, the topology and the DG, and a single fixed setting is no longer suitable for both directions.

The dual-setting directional recloser (DSDR) addresses this by giving the mid-line recloser separate pickup and time-dial settings for the forward and for the reverse direction. This project implements the method of Yousaf *et al.* [1] on the IEEE 13-node feeder in DIgSILENT PowerFactory and evaluates it.

## 1.2 Problem Statement

The problem addressed is the loss of recloser–fuse coordination caused by the change of fault-current magnitude and direction when a DG is connected. A conventional scheme that is coordinated without DG may lose the fuse-saving property in the zone between the grid and the DG. The project investigates whether the DSDR method restores the coordination in an independently built model that uses the real characteristics of the protection devices.

## 1.3 Objectives

### 1.3.1 General Objective

To build and validate a DIgSILENT PowerFactory implementation of the DSDR protection-coordination method on the IEEE 13-node test feeder, with and without distributed generation.

### **1.3.2 Specific Objectives**

1. Build and verify the IEEE 13-node feeder with its recloser and fuse protection.
2. Perform unbalanced load-flow and short-circuit studies for all fault types and phase combinations.
3. Calculate the recloser pickup currents, time dials and the fuse coefficients with the method's equations.
4. Establish the conventional recloser–fuse coordination without DG.
5. Connect the 4.05 MVA synchronous DG at node 692 and quantify the change of the fault currents and of their direction at the reclosers.
6. Classify the coordination with DG for a conventional recloser R2.
7. Design the forward and reverse settings of R2 as a DSDR and classify the coordination again and determine the operating times.
8. Verify the reclosing sequence in the time domain.
9. Study how the coordination degrades as the DG penetration rises from 0 to 100 %.
10. Compare the results with the reference paper and explain the differences (Appendix E).

## **1.4 Scope of the Study**

The study covers the IEEE 13-node feeder: unbalanced load flow, LG, LL, LLG and LLL faults at twelve fault locations, pickup and time-dial calculation, fuse coordination, the DSDR forward and reverse settings, the coordination classification with and without the DSDR, and one EMT simulation. The IEEE 34-node feeder is outside the completed work.

## **1.5 Significance of the Study**

The study gives a reproducible implementation of a protection-coordination method for feeders with DG in a tool that utilities use. It shows what the method achieves when the real relay dials and fuse sizes are respected, and it separates the contribution of the dual setting from the contribution of the fuse revision.

## **1.6 Limitations of the Study**

- The method leaves the settings of R2, the fuse sizes, the source impedance and the fault type of each operating-time case open. These were chosen; every choice and its basis is listed in Section 4.8.
- The library model of the CDG34 relay is not directional. The two setting groups are modelled as separate relay units and the direction is assigned from the fault location.
- Coordination is judged with a zero-margin criterion (the fuse must not start to melt before the fast trip). Section 5.15 shows how the result changes with a breaker time and a fuse margin.
- The DG penetration study (Section 5.16) uses one DG location (692) and keeps the protection of the design without DG at every level.
- The IEEE 34-node feeder has not been modelled.

## **1.7 Organization of the Report**

Chapter One introduces the problem and the objectives. Chapter Two reviews the protection concepts and the reference method. Chapter Three gives the method and its equations. Chapter Four describes the PowerFactory model, the DG, the protection devices and the simulation cases. Chapter Five presents the results. Chapter Six discusses the findings and their limits. Chapter Seven gives the conclusions and the remaining work. The comparison with the reference paper is in Appendix E.



# **CHAPTER TWO: LITERATURE REVIEW AND REFERENCE PAPER**

## **2.1 Overview of Overcurrent Protection**

An overcurrent device operates when the current exceeds its pickup. With an inverse-time characteristic the operating time falls as the multiple of pickup rises, so a device close to the fault clears it quickly while upstream devices act later as backup [3].

## **2.2 Distribution Network Protection**

Distribution feeders combine reclosers and fuses. A recloser distinguishes temporary from permanent faults by its fast and delayed operations; a fuse isolates a permanent fault on a lateral.

## **2.3 Overcurrent Relays**

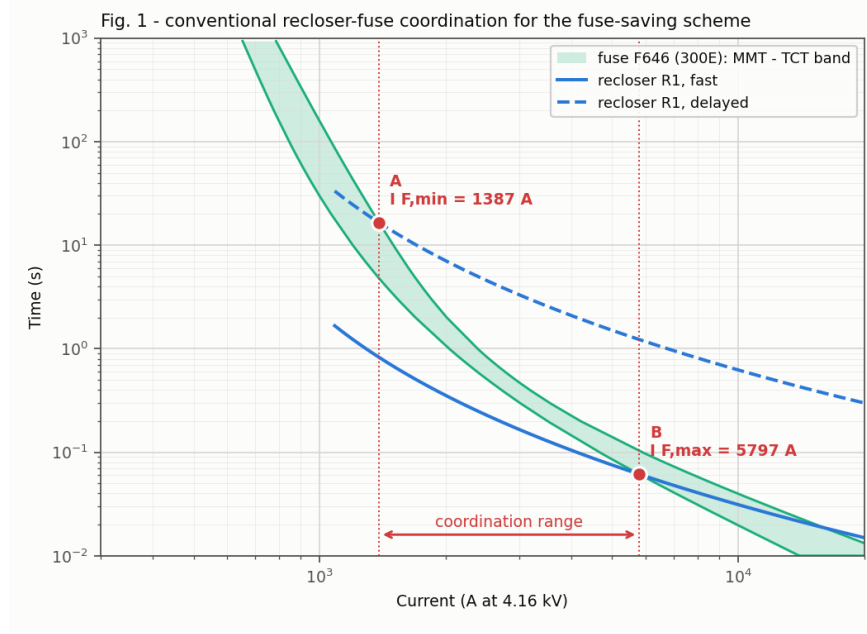
The pickup setting fixes the current above which the inverse-time characteristic is active. The time dial scales the operating time without changing the shape of the curve. Real relays offer discrete taps and a limited dial range, which is decisive in this study.

## **2.4 Directional Overcurrent Relays**

A directional element compares the current with a polarising voltage and so distinguishes the direction of the fault. It becomes necessary when a DG makes the fault current flow from the feeder towards the substation.

## **2.5 Fuse–Recloser Coordination**

For fuse saving the fast recloser curve must lie below the minimum-melting curve of the fuse, and the delayed curve above its total-clearing curve. Fuses in series are coordinated with the 75 % rule: the downstream fuse must clear within 75 % of the melting time of the upstream fuse. Figure effig:f1 shows the principle with the curves of the model.



**Figure 1:** Conventional recloser–fuse coordination: fast and delayed curve of R1 around the band of fuse F646. Coordination holds between the two crossing currents.

## 2.6 Impact of Distributed Generation on Protection Coordination

A DG raises the current through the fuses and can reverse the current through a recloser. The recloser and the fuse then no longer carry the same current, and a setting that was coordinated without DG may let the fuse melt before the fast trip [4].

## 2.7 Adaptive Protection

Adaptive protection changes the settings with the state of the network. In the DSDR the device holds two setting groups and selects one from the direction of the fault current, so no communication is needed.

## 2.8 Protection of Unbalanced Distribution Networks

The IEEE 13-node feeder [2] has single-phase, two-phase and three-phase sections and unbalanced loads. The calculations must therefore keep the phases that really exist at each node and be made per phase.

## 2.9 Review of Related Research

Naiem *et al.* [4] classify whether recloser–fuse coordination exists for each fault with DG. Yousaf *et al.* [5] give a benchmark overcurrent protection scheme, with device types and fuse sizes, for the IEEE 13-node feeder; the present work takes the fuse sizes that the reference paper does not show from that source. Kersting and Shirek [6] publish the short-circuit currents of the IEEE test feeders, which are used here to validate the fault levels.

## 2.10 Reference Paper Followed

The principal reference is M. Yousaf, A. Jalilian, K. M. Muttaqi and D. Sutanto, “An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional Recloser and Fuse Coordination in Unbalanced Distribution Networks With Distributed Generation,” *IEEE Transactions on Industry Applications*, vol. 58, no. 2, pp. 1831–1842, 2022 [1]. The project uses its equations for the forward and reverse operating times, the pickup currents, the fuse characteristic and coefficients, the series-fuse rule and the setting constraints, and its flowchart for the design sequence.

## 2.11 Research Gap

The paper reports that the DSDR restores coordination in every case, but it does not publish the settings of the DSDR, and its figures use fuse curves that differ from its own fuse equation. The gap addressed here is an independent implementation in PowerFactory that uses the real dial ranges and fuse sizes, shows which part of the improvement comes from the dual setting, and states where the method reaches the limits of the devices.

## 2.12 Summary of Literature Review

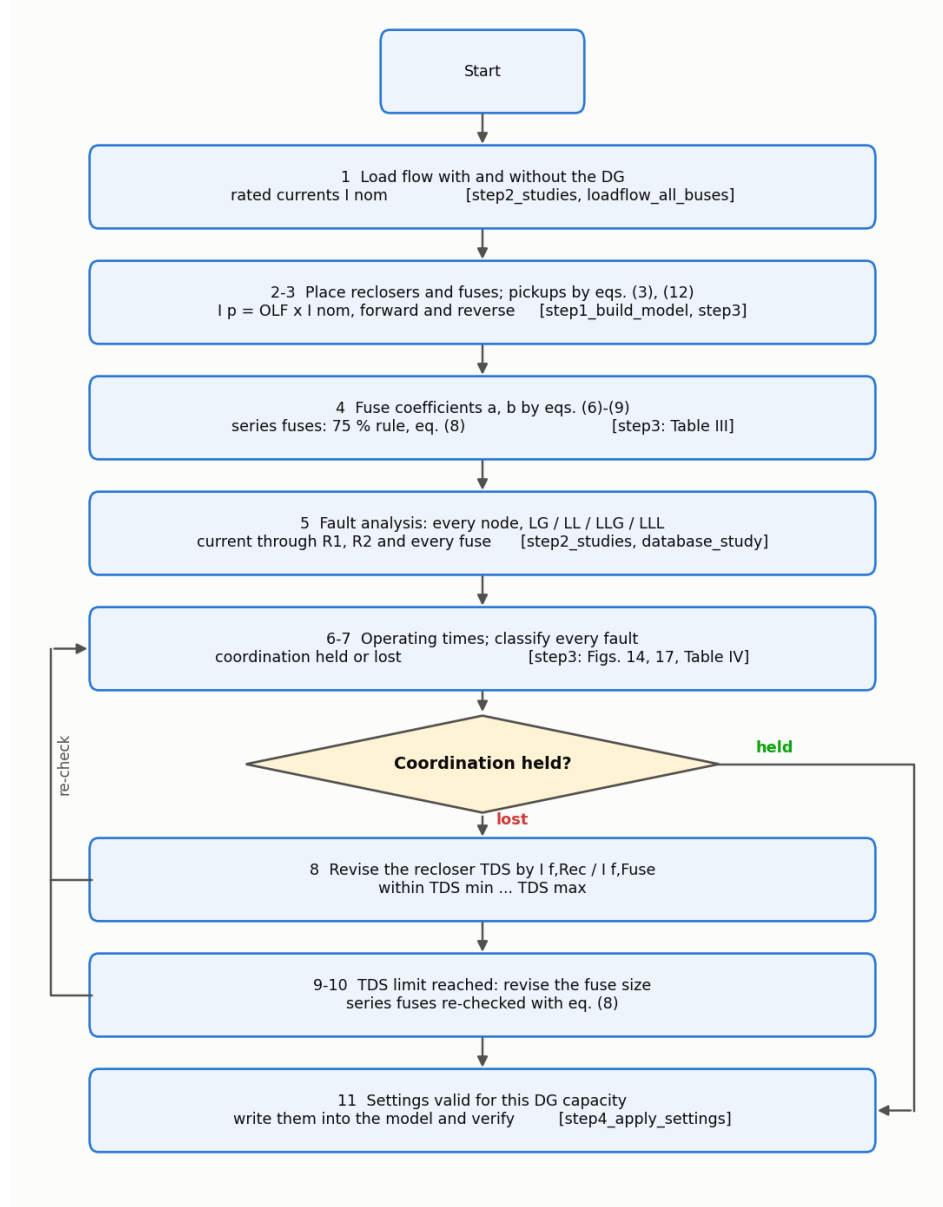
The literature establishes that a DG creates bidirectional fault currents and degrades conventional coordination, and the DSDR is a structured way to assign independent forward and reverse settings. The project adopts the method and validates it stage by stage, starting from the feeder without DG.

## **CHAPTER THREE: METHODOLOGY**

### **3.1 Research Methodology**

The work follows the flowchart of the reference paper, shown in Figure 2 as it is implemented. It has three stages: (A) the conventional scheme without DG, (B) the same settings with the DG connected, and (C) R2 as a dual-setting directional recloser with the fuse revision of the method.

Fig. 5 - the method for recloser-fuse coordination with the DSDR, with the script of each step



**Figure 2:** Flowchart of the method as implemented (Fig. 5 of the reference paper).

### 3.2 Model Setup in DIgSILENT PowerFactory

The earlier phase of the project used MATLAB/Simulink Specialized Power Systems. That model could not solve the dynamic synchronous machine reliably, used a substitute relay curve, and left several fault cases unresolved. The study was therefore rebuilt in DIgSILENT PowerFactory 2021 SP2 [7]. The feeder is the IEEE 13-node model of the DIgSILENT example library; its lines, loads and regulator taps were checked against the IEEE data. The study case

that includes the 115/4.16 kV, 5 MVA substation transformer is used, because it reproduces the IEEE short-circuit benchmark [6].

All studies are run through the PowerFactory Python interface. Table 1 lists the scripts; the command `python run_all.py` rebuilds the protection system and every result in about five minutes.

| Script                                                              | Purpose                                                                                  |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <code>step1_build_model.py</code>                                   | Places the 15 fuses and the two reclosers, sets the DG to Table I (PowerFactory)         |
| <code>step2_studies.py</code>                                       | Load flows and 273 short circuits; currents through every device (PowerFactory)          |
| <code>step3_design_and_evaluate.py</code>                           | Settings by the paper's method, classification, Tables II–IV, TCC figures                |
| <code>step4_apply_settings.py</code>                                | Writes the settings into PowerFactory and compares PowerFactory's trip times with step 3 |
| <code>step5_fig10_emt.py</code>                                     | EMT simulation of the reclosing sequence (PowerFactory)                                  |
| <code>loadflow_all_buses.py</code> , <code>database_study.py</code> | Load-flow and short-circuit database, DG out and in                                      |
| <code>make_coordination_diagrams.py</code>                          | Single setting against dual setting on selected faults                                   |

**Table 1:** Scripts of the study.

### 3.3 Load Flow Analysis

The load flow is the unbalanced three-phase AC load flow of PowerFactory. It gives the rated current of every protected branch (largest phase), the bus voltages per phase, and the current and its direction at the two reclosers with and without the DG.

### 3.4 Recloser Pickup Current

The paper specifies an overload factor of 1.25. The pickup is 1.25 times the rated current of the protected branch, eq. (3), and is then set on the nearest available tap of the relay.

### 3.5 Short-Circuit Analysis

Short circuits are calculated with the *complete method* of PowerFactory, which starts from the pre-fault load flow. LG, LL, LLG and LLL faults are applied at every node for every phase combination that exists there, with the DG out and in. The maximum fault current of a branch is a bolted fault at its nearest node; the minimum is an LG fault through  $3\ \Omega$  at its farthest node, as in the paper. The current through R1, R2, every fuse and every branch of Table II is stored for each fault.

### 3.6 Mathematical Formulation of the DSDR

The equations below are those of the reference paper; the numbering follows the paper.

#### 3.6.1 Forward Fast and Delayed Operating Time

$$t_{k,l,F}^{fw} = TDS_{k,F}^{fw} \left[ \frac{A_k^{fw}}{\left( I_{f,l}/I_{p,k}^{fw} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right] \quad (1)$$

$$t_{k,l,D}^{fw} = TDS_{k,D}^{fw} \left[ \frac{A_k^{fw}}{\left( I_{f,l}/I_{p,k}^{fw} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right] \quad (2)$$

The fast and the delayed operation of recloser  $k$  for a fault at location  $l$  use the same curve with two time dials.  $I_{f,l}$  is the fault current through the recloser and  $I_p$  its pickup.

#### 3.6.2 Forward Pickup Current and Constraints

$$I_{p,k}^{fw} = OLF \times I_{nom,k}^{fw} \quad (3)$$

$$TDS_{k,min} < TDS_{k,F}^{fw}, TDS_{k,D}^{fw} < TDS_{k,max} \quad (4)$$

$$I_{p,k,min} < I_{p,k}^{fw} < I_{p,k,max} \quad (5)$$

The pickup lies above the load current and below the minimum fault current of the zone; the dials must stay inside the range of the device.

#### 3.6.3 Fuse Characteristic and Coefficient

$$\log t_{fuse,i} = a_i \log I_{f,i} + b_i \quad (6)$$

$$b_i = \log \left[ \frac{t_{k,F}^{fw} + t_{k,D}^{fw}}{2} \right] - a_i \log I_{f,i} \quad (7)$$

with  $a_i = -1.8$ . Equation (7) places a single fuse midway between the fast and the delayed time at the calibration current.

### 3.6.4 Series Fuses

$$t_{TCT,i} \leq 0.75 t_{MMT,i+1} \quad (8)$$

$$b_i = \log \left[ t_{k,F}^{fw} + \frac{i \left( t_{k,D}^{fw} - t_{k,F}^{fw} \right)}{z + 1} \right] - a_i \log I_{f,i} \quad (9)$$

where  $z$  is the number of series fuses in the fault path and  $i = 1$  is the fuse closest to the fault. Equation (9) divides the window between the two recloser times among the series fuses.

### 3.6.5 Reverse Operating Times, Pickup and Constraints

$$t_{k,m,F}^{rv} = TDS_{k,F}^{rv} \left[ \frac{A_k^{rv}}{\left( I_{f,m}/I_{p,k}^{rv} \right)^{n_k^{rv}} - 1} + B_k^{rv} \right] \quad (10)$$

$$t_{k,m,D}^{rv} = TDS_{k,D}^{rv} \left[ \frac{A_k^{rv}}{\left( I_{f,m}/I_{p,k}^{rv} \right)^{n_k^{rv}} - 1} + B_k^{rv} \right] \quad (11)$$

$$I_{p,k}^{rv} = OLF \times I_{nom,k}^{rv} \quad (12)$$

$$TDS_{k,min} < TDS_{k,F}^{rv}, TDS_{k,D}^{rv} < TDS_{k,max} \quad (13)$$

$$I_{p,k,min} < I_{p,k}^{rv} < I_{p,k,max} \quad (14)$$

Here  $m$  is a fault location upstream of the recloser. The reverse current is the contribution of the DG, and the reverse pickup is based on the reverse load current that flows when the DG exports towards the substation.

## 3.7 Device Characteristics Used in the Model

The paper names the devices but does not print their curve constants. The model uses the curves of the PowerFactory library for these devices.



### 3.7.1 R1: GE IAC77B801A

The extremely inverse IAC curve is

$$t = TDS \left[ 0.004 + \frac{0.6379}{M - 0.62} + \frac{1.7872}{(M - 0.62)^2} + \frac{0.2461}{(M - 0.62)^3} \right], \quad M = \frac{I_f}{I_p}, \quad (15)$$

valid from  $M = 1.5$  to  $M = 40$ . For 4219 A, for example, it gives 0.096 s (fast, TDS 0.5) and 1.926 s (delayed, TDS 10).

### 3.7.2 R2: GE/Alstom CDG34

The CDG extremely inverse characteristic is a manufacturer's table of time against  $M = I_f/I_s$  for time multipliers from 0.1 to 1.0. It starts at twice the plug setting and is interpolated on log-log axes:

$$t = t_1 \left( \frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(M/M_1)}{\ln(M_2/M_1)}. \quad (16)$$

The table is not defined below a multiplier of 0.1, so 0.1 is the fastest setting of this relay.

### 3.7.3 Fuses

The fuses are Gould-Shawmut A055C (5.5 kV, E-rated) with their minimum-melting and total-clearing curves. The fuse times are read from these curves at the current through the fuse with the same interpolation as eq. (16). The straight line of eq. (6) is used only to compute the coefficients of Table III.

## 3.8 Coordination Criteria

A fault cell is classified as *coordination exists* when all of the following hold:

$$t_{MMT}(I_{fuse}) > t_F \quad \text{for every recloser that feeds the fault through the fuse,} \quad (17)$$

$$t_{TCT}(I_{fuse}) < t_D \quad \text{for the same reclosers,} \quad (18)$$

together with the 75 % rule of eq. (8) for series fuses and a margin of at least 10 cycles between the delayed curves of R2 and R1. A cell (node and fault type) is held only if every phase combination of that fault type is held. The fuse and each recloser are evaluated at their own currents, because with DG they differ.

### 3.9 Design Procedure

The steps of Figure 2 are applied as follows.

1. Load flow without DG: rated currents (Table II) and pickups, eq. (3).
2. Short-circuit study: minimum and maximum fault currents (Table II).
3. Recloser dials: R1 as given in the paper; R2 fast at the lowest dial, R2 delayed at the highest dial that stays 10 cycles below R1.
4. Fuse coefficients, eqs. (7) and (9) (Table III), and check of the installed fuses.
5. Classification without DG; fuse revision until every cell is held.
6. DG connected, settings unchanged: classification (Fig. 14).
7. Reverse pickup of R2, eq. (12), and reverse dials.
8. Revision of the dials (step 8 of the paper) and of the fuse sizes (step 9) until the cells are held.
9. Classification with the DSDR (Fig. 17) and operating times (Table IV).
10. Verification: PowerFactory's own relay and fuse models, and the EMT simulation.

## CHAPTER FOUR: SYSTEM MODELING AND SIMULATION SETUP

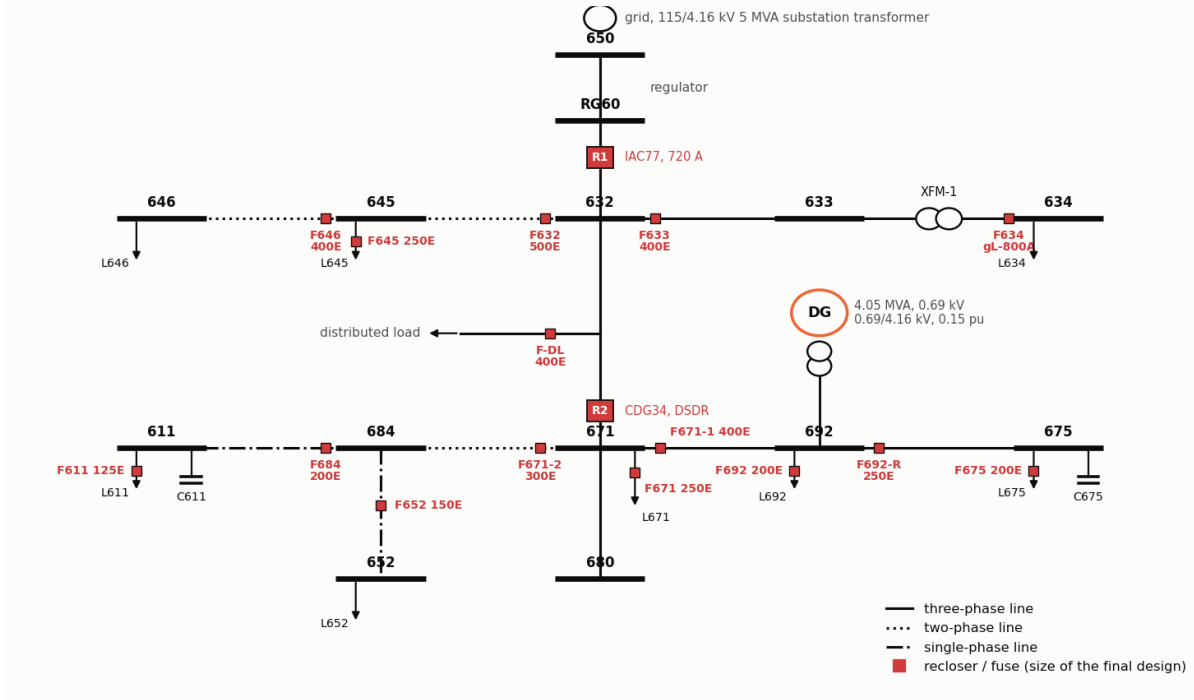
### 4.1 DIgSILENT PowerFactory Environment

The model is the project *IEEE13 DSDR Fuse Coordination* in PowerFactory 2021 SP2. Load flows use the unbalanced AC method, short circuits the complete method, and the time-domain study the EMT simulation with a step of 0.1 ms. The studies are run from Python 3.9 in engine mode; the same sequence can be started inside PowerFactory from one script object, which prints the report in the output window.

### 4.2 IEEE 13-Node Feeder Model

Figure 3 shows the feeder with the protection devices of the reference paper. The feeder has the voltage regulator at the substation, overhead and cable sections, the 4.16/0.48 kV transformer XFM-1, spot and distributed loads, and two capacitors. The middle part of the distributed load on line 632–671 is placed behind a short lateral (node DL) so that a fault there lies below its fuse, as drawn in the paper.

Fig. 6 - IEEE 13-node test feeder with the protective devices of this study



**Figure 3:** IEEE 13-node feeder with the reclosers R1 and R2, the fuses and the DG at node 692.

### 4.3 Source and Feeder

The source is an external grid at 115 kV behind the 5 MVA, 115/4.16 kV substation transformer ( $u_k = 8.06\%$ ). With this source the fault currents agree with the IEEE benchmark [6] within about 2 % at every node. The alternative study case with a stiff 4.16 kV source overstates the fault levels by 20 to 65 % and is not used. The regulator taps are fixed at +10, +8 and +11.

### 4.4 Distributed Generation Model

The DG is a 4.05 MVA, 0.69 kV synchronous machine connected to node 692 through a 0.69/4.16 kV transformer with a leakage reactance of 0.15 p.u. It is modelled with the standard synchronous-machine model of PowerFactory, round rotor, with the short-circuit data of Table 2 (values read back from the model). In the load flow the machine delivers 3.24 MW and controls its terminal voltage to 1.0 p.u.

| Parameter                                      | Symbol     | Value  | Unit |
|------------------------------------------------|------------|--------|------|
| Leakage reactance                              | $X_l$      | 0.05   | pu   |
| Stator resistance                              | $R_a$      | 0.0014 | pu   |
| d-axis synchronous reactance                   | $X_d$      | 1.4    | pu   |
| d-axis transient reactance                     | $X'_d$     | 0.231  | pu   |
| d-axis subtransient reactance                  | $X''_d$    | 0.118  | pu   |
| d-axis transient open-circuit time constant    | $T'_{d0}$  | 5.5    | s    |
| d-axis subtransient open-circuit time constant | $T''_{d0}$ | 0.05   | s    |
| q-axis synchronous reactance                   | $X_q$      | 1.372  | pu   |
| q-axis transient reactance                     | $X'_q$     | 0.8    | pu   |
| q-axis subtransient reactance                  | $X''_q$    | 0.118  | pu   |
| q-axis transient open-circuit time constant    | $T'_{q0}$  | 1.25   | s    |
| q-axis subtransient open-circuit time constant | $T''_{q0}$ | 0.19   | s    |
| Mechanical starting time                       | $M = 2H$   | 1.5    | s    |
| DG rating                                      | $S_n$      | 4.05   | MVA  |
| DG voltage                                     | $U_n$      | 0.69   | kV   |
| Step-up transformer, HV side                   | $U_{HV}$   | 4.16   | kV   |
| Step-up transformer leakage reactance          | $x_T$      | 0.15   | pu   |

**Table 2:** Parameters of the synchronous DG in the PowerFactory model.

## 4.5 Recloser Model

R1 is at the feeder head (RG60–632). R2 is on line 632–671 at the 671 end; it is the DSDR, because with the DG at 692 it carries forward current for faults at 671 and beyond and reverse current for faults towards 632. Table 3 gives the settings. The forward and the reverse group of R2 are four relay units of the CDG34 library type on two current transformers.

| Recloser   | Relay, CT            | $1.25 I_{nom}$ | Setting            | Dial fast / delayed |
|------------|----------------------|----------------|--------------------|---------------------|
| R1 forward | GE IAC77B801A, 900/5 | 734.6 A        | tap 4 A = 720 A    | TDS 0.5 / 10        |
| R2 forward | CDG34, 1000/5        | 587.5 A        | plug 300 A / 600 A | TMS 0.1 / 1.0       |
| R2 reverse | CDG34, 500/5         | 315.3 A        | plug 150 A / 300 A | TMS 0.1 / 1.0       |

**Table 3:** Recloser settings. For R2 the fast plug is half and the delayed plug equal to  $1.25 I_{nom}$ , because the CDG curve starts at twice the plug setting.

## 4.6 Fuse Model

Table 4 lists the fifteen fuses. The starting size is the size drawn in the figures of the reference paper where a figure shows the fuse (the fuse bands were digitised and matched to the A055C library curves within 2 %), otherwise the size of the benchmark scheme [5]. The last two columns are the sizes after the fuse revision of the method without DG and with the DSDR.

| Fuse   | Location                            | Start   | No DG   | DSDR        | Source of the starting size      |
|--------|-------------------------------------|---------|---------|-------------|----------------------------------|
| F632   | lateral 632–645, at 632             | 400E    | 400E    | <b>500E</b> | Paper, Figs. 12, 15, 16          |
| F633   | lateral 632–633, at 632             | 250E    | 250E    | <b>400E</b> | Benchmark scheme [5],<br>Table V |
| F634   | XFM-1, 0.48 kV side                 | gL-800A | gL-800A | gL-800A     | Benchmark scheme [5],<br>Table V |
| F645   | load at 645                         | 250E    | 250E    | 250E        | Benchmark scheme [5],<br>Table V |
| F646   | line 645–646, at 645                | 300E    | 300E    | <b>400E</b> | Paper, Figs. 11, 15, 16          |
| F-DL   | distributed-load lateral on 632–671 | 250E    | 250E    | <b>400E</b> | Benchmark scheme [5],<br>Table V |
| F671   | load at 671                         | 250E    | 250E    | 250E        | Benchmark scheme [5],<br>Table V |
| F671-1 | 671–692 (switch), at 671            | 300E    | 300E    | <b>400E</b> | Paper, Fig. 9                    |
| F692   | load at 692                         | 200E    | 200E    | 200E        | Benchmark scheme [5],<br>Table V |
| F692-R | cable 692–675, at 692               | 250E    | 200E    | <b>250E</b> | Paper, Fig. 9                    |
| F675   | load at 675                         | 200E    | 200E    | 200E        | Benchmark scheme [5],<br>Table V |
| F671-2 | lateral 671–684, at 671             | 300E    | 300E    | 300E        | Paper, Fig. 8                    |
| F684   | line 684–611, at 684                | 200E    | 200E    | 200E        | Paper, Fig. 8                    |
| F611   | load at 611                         | 125E    | 125E    | 125E        | Benchmark scheme [5],<br>Table V |
| F652   | cable 684–652, at 684               | 150E    | 150E    | 150E        | Benchmark scheme [5],<br>Table V |

**Table 4:** Fuses and their sizes (A055C, E-rated). Bold: changed by the fuse revision with the DG.

## 4.7 Fault Model

Faults are applied at the nodes 632, 633, 634, 645, 646, DL, 671, 692, 675, 680, 684, 611 and 652. Only the fault types that the phases of a node allow are applied, for example LL and LLG on phases b–c at 645 and 646, and LG only at 611 and 652. This gives 39 node/fault-type cells. The faults of the paper’s figures are added with their fault impedances.

## 4.8 Measurement and Data Acquisition

For every fault the phase currents through R1, R2 and each fuse are read from the short-circuit result. The operating current of a device is its largest phase current. The forward direction is away from the grid; R2 sits at the 671 end of its line, where PowerFactory counts the current positive into the line, so its values are multiplied by  $-1$ . The following values are not given in the paper and were chosen:

- the source (IEEE benchmark substation transformer) and the complete short-circuit method;
- the tap and the current transformer of R1 (720 A reproduces the paper's worked example);
- all settings of R2 (Table 3), derived from eqs. (3) and (12) and from the curve starts visible in the paper's figures;
- the fault behind each row of Table IV: bolted LLL, LL at two-phase nodes, LG at single-phase nodes;
- the reclosing dead time (0.2 s, read from the paper's Fig. 10) and a breaker time of three cycles in the EMT study.

## 4.9 Simulation Cases

| Case           | DG       | Faults                                                   | Purpose                                          |
|----------------|----------|----------------------------------------------------------|--------------------------------------------------|
| Load flow      | out / in | –                                                        | Rated currents, voltages, direction at R1 and R2 |
| Maximum faults | out / in | All nodes, LG/LL/LLG/LLL, all phase combinations, bolted | Classification, Tables II–IV                     |
| Minimum faults | out / in | LG through $3\ \Omega$ at all nodes                      | $I_{f,min}$ , pickup check                       |
| Figure faults  | in       | The faults of Figs. 8–13, 15, 16 of the paper            | TCC comparison                                   |
| EMT            | out / in | LL a–c at 684 through $0.2\ \Omega$                      | Reclosing sequence (Fig. 10)                     |
| Penetration    | 0–100 %  | LLL at 633, LL at 684, all nodes                         | CTI, fault levels and pickups against DG size    |

**Table 5:** Simulation cases.



## CHAPTER FIVE: RESULTS OBTAINED

### 5.1 Load-Flow Results

Table 6 gives the bus voltages with and without the DG. Without DG the lowest voltage is 0.971 p.u. (phase c at 611). The DG raises the voltages of the loaded phases a and c at node 671 and beyond by about 0.03 to 0.05 p.u.

| Bus  | Without DG (p.u.) |        |        | With DG (p.u.) |        |        |
|------|-------------------|--------|--------|----------------|--------|--------|
|      | A                 | B      | C      | A              | B      | C      |
| 650  | 0.9981            | 1.0170 | 0.9979 | 1.0084         | 1.0200 | 1.0057 |
| RG60 | 1.0605            | 1.0679 | 1.0665 | 1.0715         | 1.0710 | 1.0748 |
| 632  | 1.0187            | 1.0616 | 1.0142 | 1.0482         | 1.0640 | 1.0361 |
| 633  | 1.0158            | 1.0588 | 1.0124 | 1.0454         | 1.0611 | 1.0344 |
| 634  | 0.9918            | 1.0406 | 0.9936 | 1.0221         | 1.0430 | 1.0161 |
| 645  | –                 | 1.0540 | 1.0106 | –              | 1.0563 | 1.0325 |
| 646  | –                 | 1.0522 | 1.0085 | –              | 1.0546 | 1.0304 |
| DL   | 1.0028            | 1.0666 | 0.9939 | 1.0407         | 1.0685 | 1.0221 |
| 671  | 0.9877            | 1.0722 | 0.9753 | 1.0334         | 1.0738 | 1.0094 |
| 692  | 0.9877            | 1.0722 | 0.9753 | 1.0334         | 1.0738 | 1.0094 |
| 675  | 0.9809            | 1.0767 | 0.9729 | 1.0271         | 1.0781 | 1.0072 |
| 680  | 0.9877            | 1.0722 | 0.9753 | 1.0334         | 1.0738 | 1.0094 |
| 684  | 0.9859            | –      | 0.9732 | 1.0315         | –      | 1.0074 |
| 652  | 0.9811            | –      | –      | 1.0265         | –      | –      |
| 611  | –                 | –      | 0.9712 | –              | –      | 1.0055 |

**Table 6:** Bus voltages from the unbalanced load flow, without and with the DG.

Table 7 shows the load current at the two reclosers. With the DG the current through R2 reverses: 252.2 A flows from 671 towards 632. This reverse load current is the basis of the reverse pickup, eq. (12).

| Device | Without DG (A) | Direction | With DG (A) | Direction |
|--------|----------------|-----------|-------------|-----------|
| R1     | +587.7         | forward   | +284.9      | forward   |
| R2     | +470.0         | forward   | −252.2      | reverse   |

**Table 7:** Load current at the reclosers (largest phase; + is forward, away from the grid).

## 5.2 Branch Currents and Fault Levels (Table II)

Table 8 gives the rated current of every protected branch from the unbalanced load flow, the minimum fault current (LG through  $3\ \Omega$  at the farthest node of the branch's section) and the maximum fault current (bolted, nearest node). The largest fault current at the feeder head is 4.73 kA; the fault levels agree with the IEEE 13-node short-circuit benchmark within about 2 %.

| Branch       | $I_{nom}$ (A) | $I_{f,min}$ (kA) | $I_{f,max}$ (kA) |
|--------------|---------------|------------------|------------------|
| RG60–632     | 587.7         | 1.07             | 4.73             |
| 632–633      | 81.0          | 0.76             | 4.1              |
| 632–645      | 143.1         | 0.73             | 3.42             |
| 632–671      | 470.0         | 0.88             | 3.39             |
| XFM1-HV side | 81.0          | 0.08             | 1.81             |
| XFM1-LV side | 706.4         | 0.71             | 15.67            |
| 645–646      | 66.1          | 0.73             | 3.11             |
| 671–692      | 229.7         | 0.74             | 3.39             |
| 671–684      | 71.1          | 0.66             | 2.66             |
| 671–680      | 0.0           | 0.64             | 2.93             |
| 692–675      | 205.9         | 0.74             | 3.15             |
| 684–611      | 71.1          | 0.68             | 1.86             |
| 684–652      | 63.0          | 0.66             | 1.87             |

**Table 8:** Rated, minimum and maximum fault current of the protected branches without DG. XFM1 HV side: 4.16 kV currents for a fault on the 0.48 kV side.

## 5.3 Recloser Pickup Currents and Settings

With the rated currents of Table 8, eq. (3) gives 734.6 A for R1 and 587.5 A for R2 forward. The settings actually applied are those of Table 3. Without DG both pickups are below the lowest minimum fault current of their zones (1074 A and 879 A).

## 5.4 Short-Circuit Results With and Without DG

Table 9 gives the largest fault current at each node and the current through the two reclosers. The DG raises the fault current at every node, by 38 % at 632 and by 64 % at 671. For faults at 632, 633, 645, 646 and DL the current through R2 becomes negative: R2 then carries only the DG contribution, 1.1 to 2.0 kA, in the reverse direction, while R1 still carries the grid contribution.

| Bus | Fault   | $I_{f,max}$ at the fault (A) |       | Through R1 (A) |       | Through R2 (A) |       |
|-----|---------|------------------------------|-------|----------------|-------|----------------|-------|
|     |         | no DG                        | DG    | no DG          | DG    | no DG          | DG    |
| 632 | LLL ABC | 4735                         | 6526  | +4735          | +4734 | +0             | −1813 |
| 633 | LLL ABC | 4102                         | 5426  | +4162          | +3970 | +66            | −1500 |
| 634 | LLL ABC | 15671                        | 17608 | +2034          | +1611 | +298           | −618  |
| 645 | LLG BC  | 3418                         | 4341  | +3528          | +3330 | +478           | −1262 |
| 646 | LLG BC  | 3112                         | 3854  | +3192          | +3000 | +475           | −1135 |
| 671 | LLL ABC | 3390                         | 5564  | +3413          | +3412 | +3390          | +3391 |
| 692 | LLL ABC | 3390                         | 5564  | +3413          | +3412 | +3390          | +3391 |
| 675 | LLL ABC | 3151                         | 5022  | +3202          | +3090 | +3174          | +3050 |
| 680 | LLL ABC | 2930                         | 4460  | +3006          | +2789 | +2976          | +2757 |
| 684 | LLG AC  | 2655                         | 4107  | +2836          | +2669 | +2744          | +2651 |
| 652 | LG A    | 1874                         | 2239  | +2094          | +1795 | +2059          | +1761 |
| 611 | LG C    | 1857                         | 2236  | +2084          | +1814 | +1991          | +1718 |
| DL  | LLL ABC | 3946                         | 5902  | +3959          | +3954 | +1             | −1972 |

**Table 9:** Maximum fault current and recloser currents without and with the DG (+ forward, − reverse).

## 5.5 Fuse Coefficients (Table III)

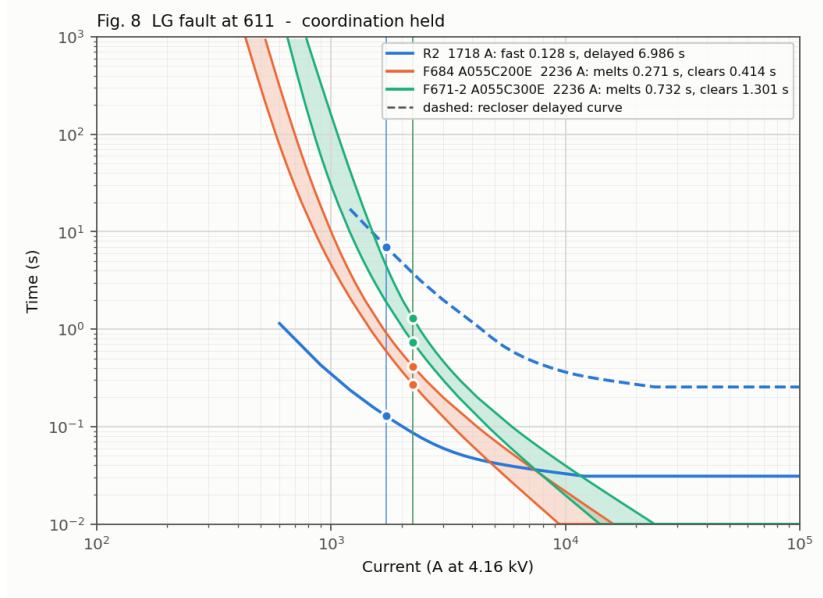
Table 10 gives the fuse coefficients of eq. (9), with  $i = 1$  for the fuse closest to the fault. The last columns give the installed fuse and its melting time at the same current; it is much shorter than the time the equation asks for, because real fuse curves are steeper than the slope  $-1.8$ .

| Fuse   | $I_f$ (A) | $i/z$ | $t_F$ (s) | $t_D$ (s) | $t_{fuse}$ (s) | $b_i$ | Size    | $t_{MMT}$ (s) |
|--------|-----------|-------|-----------|-----------|----------------|-------|---------|---------------|
| F632   | 3418      | 2/2   | 0.133     | 2.670     | 1.824          | 6.62  | 400E    | 0.512         |
| F633   | 4102      | 1/1   | 0.100     | 2.008     | 1.054          | 6.53  | 250E    | 0.102         |
| F634   | 15671     | 1/2   | 0.439     | 8.772     | 3.217          | 8.06  | gL-800A | 0.021         |
| F645   | 3418      | 1/2   | 0.133     | 2.670     | 0.979          | 6.35  | 250E    | 0.158         |
| F646   | 3112      | 1/2   | 0.156     | 3.115     | 1.142          | 6.35  | 300E    | 0.273         |
| F-DL   | 3946      | 1/1   | 0.106     | 2.130     | 1.118          | 6.52  | 250E    | 0.112         |
| F671   | 3390      | 1/1   | 0.054     | 1.601     | 0.827          | 6.27  | 250E    | 0.161         |
| F671-1 | 3390      | 3/3   | 0.054     | 1.601     | 1.214          | 6.44  | 300E    | 0.218         |
| F692   | 3390      | 1/2   | 0.054     | 1.601     | 0.569          | 6.11  | 200E    | 0.094         |
| F692-R | 3151      | 2/3   | 0.057     | 1.829     | 0.943          | 6.27  | 200E    | 0.112         |
| F675   | 3151      | 1/3   | 0.057     | 1.829     | 0.500          | 6.00  | 200E    | 0.112         |
| F671-2 | 2655      | 3/3   | 0.069     | 2.572     | 1.946          | 6.45  | 300E    | 0.428         |
| F684   | 1857      | 2/3   | 0.113     | 5.778     | 2.946          | 6.35  | 200E    | 0.465         |
| F611   | 1857      | 1/3   | 0.113     | 5.778     | 1.530          | 6.07  | 125E    | 0.120         |
| F652   | 1874      | 1/2   | 0.112     | 5.660     | 1.961          | 6.18  | 150E    | 0.207         |

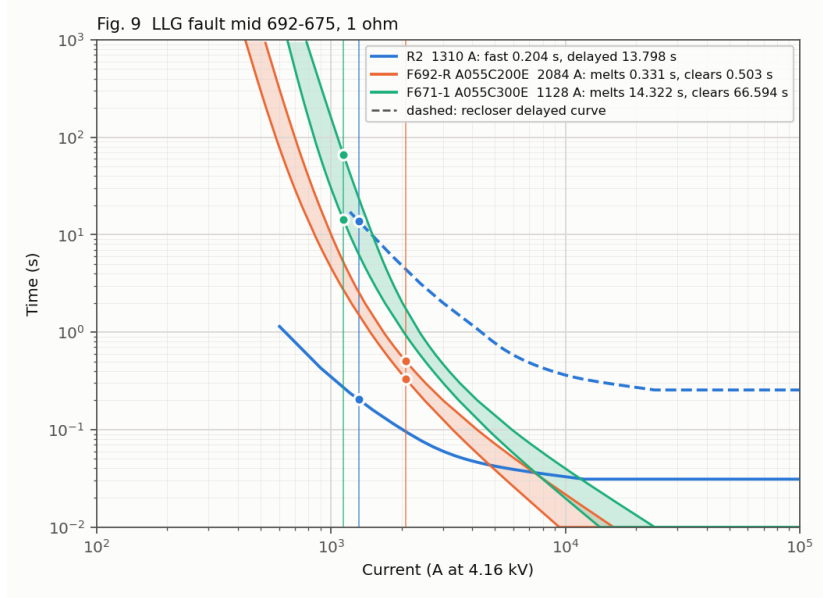
**Table 10:** Fuse coefficients, eq. (9),  $a_i = -1.8$ . Size and  $t_{MMT}$ : fuse of the design without DG at the same current.

## 5.6 Conventional Coordination Without DG

With the starting fuse sizes 35 of the 39 cells are coordinated. The four lost cells are the faults at 675, where the series pair F692-R (250E) and F671-1 (300E) breaks the 75 % rule. The fuse revision of the method changes F692-R to 200E, after which all 39 cells are coordinated. Figures 4 and 5 show two cases: R2's fast curve lies below the fuses and its delayed curve above them.



**Figure 4:** LG fault at 611: R2 forward, fuses F684 and F671-2.



**Figure 5:** LLG fault at the middle of 692–675 through 1  $\Omega$ .

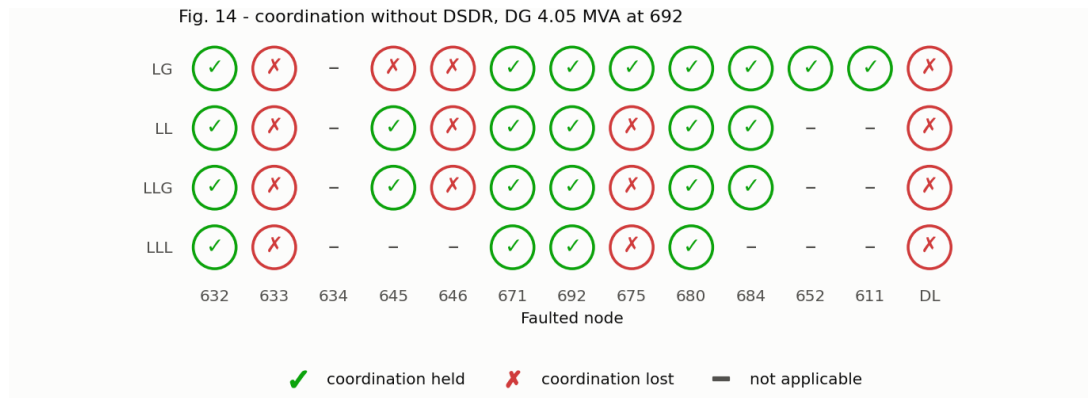
## 5.7 Coordination With DG and a Conventional R2 (Fig. 14)

With the DG connected and the settings unchanged, coordination is held in 24 of 39 cells (Table 11, Figure 6). The lost cells are at 633, 645, 646, DL and 675: the fuse now carries the grid and the DG current and melts before the fast trip, and for some LG faults the single-setting R2

does not pick up the reverse DG current at all. At 633 and DL the fuses have almost no margin even without DG, so every fault type is lost there.

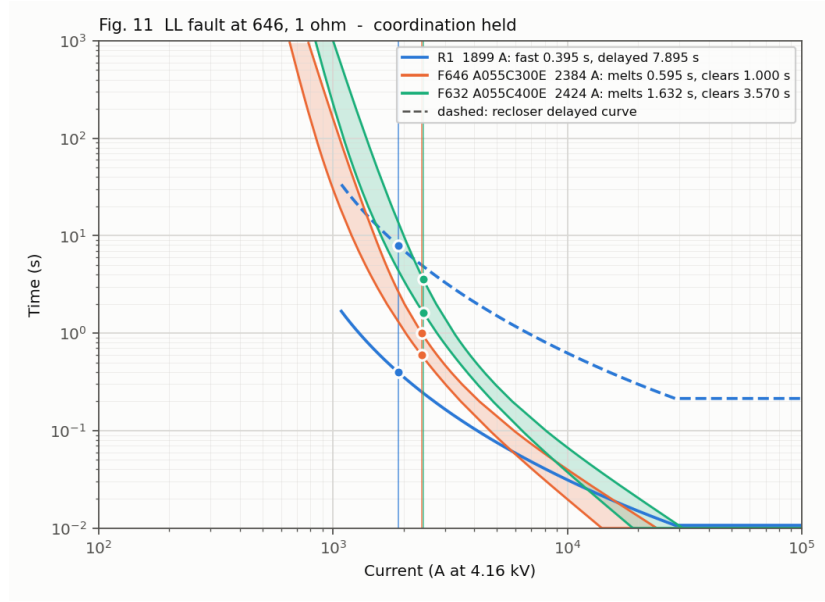
| Fault | 632 | 633 | 645 | 646 | DL | 671 | 692 | 675 | 680 | 684 | 611 | 652 |
|-------|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|
| LG    | ✓   | ×   | ×   | ×   | ×  | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   |
| LL    | ✓   | ×   | ✓   | ×   | ×  | ✓   | ✓   | ×   | ✓   | ✓   | –   | –   |
| LLG   | ✓   | ×   | ✓   | ×   | ×  | ✓   | ✓   | ×   | ✓   | ✓   | –   | –   |
| LLL   | ✓   | ×   | –   | –   | ×  | ✓   | ✓   | ×   | ✓   | –   | –   | –   |

**Table 11:** Coordination status with DG and a conventional R2. ✓ exists, × does not exist, – not applicable.

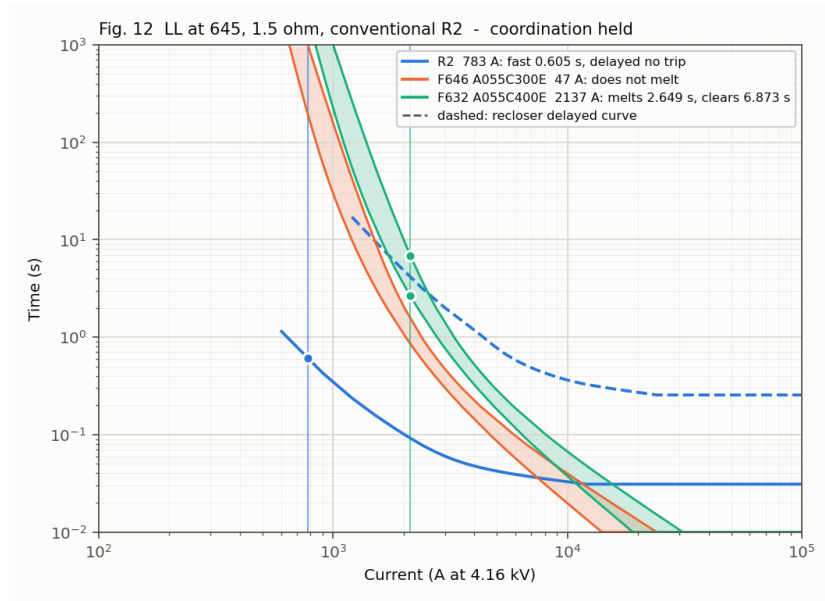


**Figure 6:** Coordination status with DG, without the DSDR.

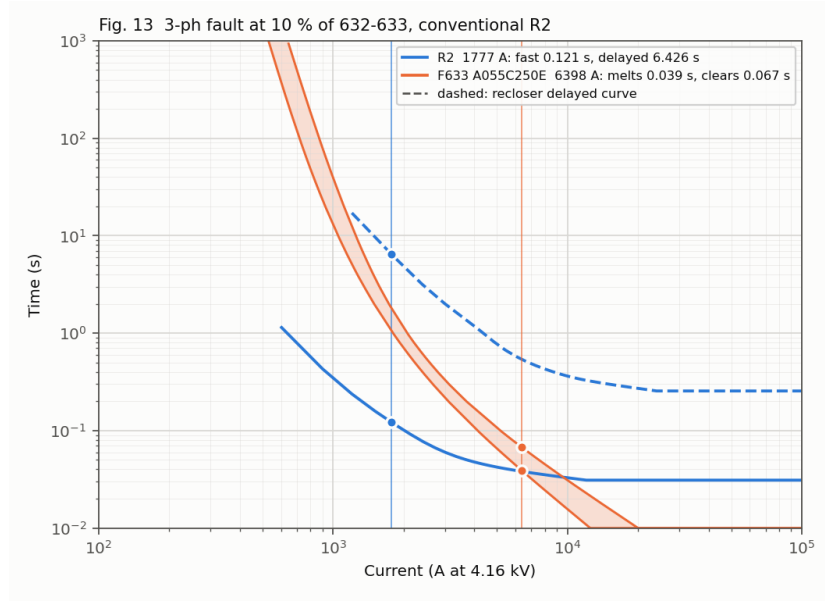
Figures 7 to 9 show three cases with the conventional R2.



**Figure 7:** LL fault at 646 through 1  $\Omega$ : R1 with F646 and F632.



**Figure 8:** LL fault at 645 through 1.5  $\Omega$ , conventional R2.



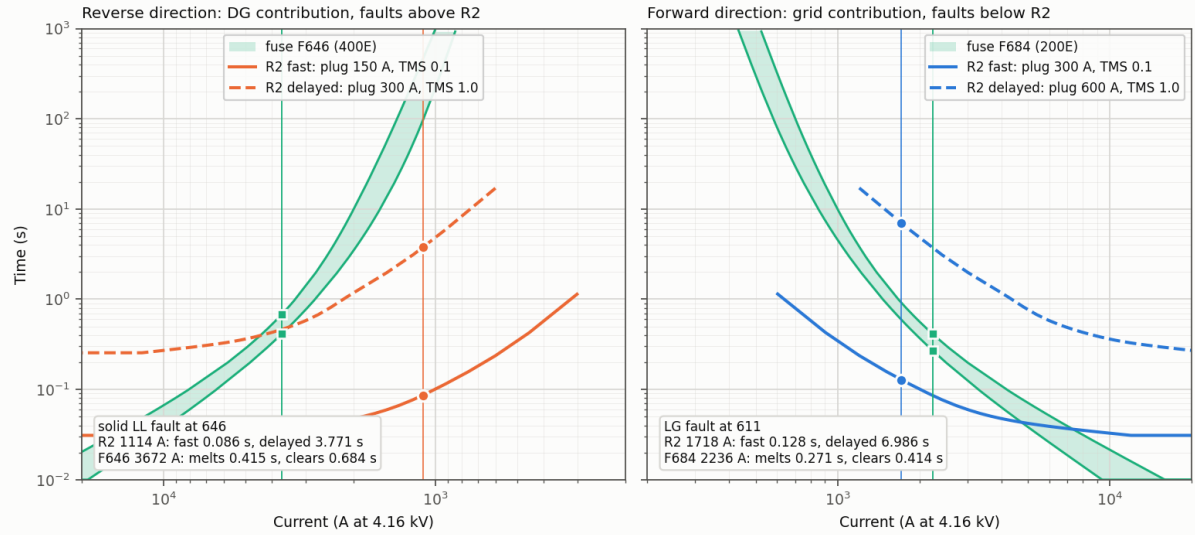
**Figure 9:** Three-phase fault at 10 % of line 632–633, conventional R2.

## 5.8 DSDR Settings and Fuse Revision

The reverse load current of R2 is 252.2 A, so eq. (12) gives a reverse pickup of 315.3 A. The reverse group is set on a 500/5 current transformer with plugs of 150 A (fast) and 300 A (delayed) and the same time multipliers 0.1 and 1.0 (Figure 10). Step 8 of the method (revising the dials) could not improve any lost case, because R1 is already at both ends of the IAC dial range (0.5 and 10) and R2 at both ends of the CDG table (0.1 and 1.0). Step 9 (revising the fuses) was therefore needed: F632 to 500E; F633, F646, F-DL and F671-1 to 400E; F692-R to 250E (Table 4).



Fig. 3 - time-current characteristics of the dual-setting recloser R2: reverse and forward setting



**Figure 10:** The two setting groups of R2: reverse (DG contribution, faults towards 632) and forward (grid contribution, faults beyond 671), each with one fault of the model.

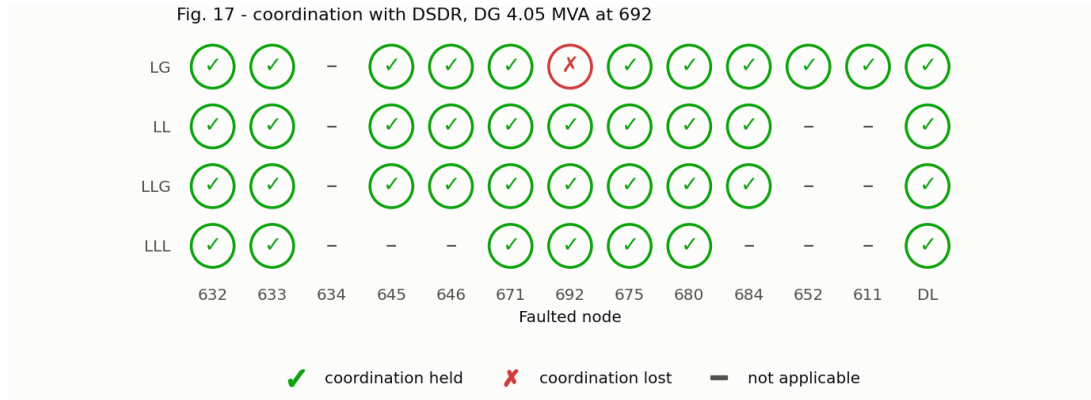
## 5.9 Coordination With the DSDR (Fig. 17)

With R2 as a DSDR and the revised fuses, coordination is held in 38 of 39 cells (Table 12, Figure 11). With the same fuses and a single-setting R2 only 31 cells are held. The dual setting itself therefore restores seven cells: all four fault types at 633, and the LG faults at 645, 646 and DL.

The cell that remains lost is the LG fault at 692. F671-1 must be 400E to stay selective with F692-R, but it then clears after R2's delayed trip (7.6 s against 4.0 s), and R2's delayed dial is already at its maximum.

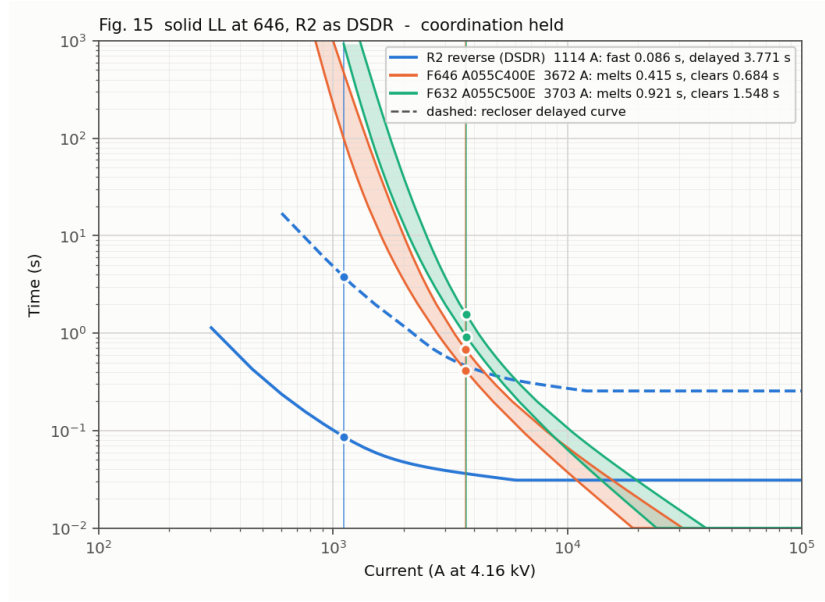
| Fault, R2           | 632 | 633 | 645 | 646 | DL | 671 | 692 | 675 | 680 | 684 | 611 | 652 |
|---------------------|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|
| LG, single setting  | ✓   | ×   | ×   | ×   | ×  | ✓   | ×   | ✓   | ✓   | ✓   | ✓   | ✓   |
| LG, dual setting    | ✓   | ✓   | ✓   | ✓   | ✓  | ✓   | ×   | ✓   | ✓   | ✓   | ✓   | ✓   |
| LL, single setting  | ✓   | ×   | ✓   | ✓   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | —   | —   |
| LL, dual setting    | ✓   | ✓   | ✓   | ✓   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | —   | —   |
| LLG, single setting | ✓   | ×   | ✓   | ✓   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | —   | —   |
| LLG, dual setting   | ✓   | ✓   | ✓   | ✓   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | —   | —   |
| LLL, single setting | ✓   | ×   | —   | —   | ✓  | ✓   | ✓   | ✓   | ✓   | —   | —   | —   |
| LLL, dual setting   | ✓   | ✓   | —   | —   | ✓  | ✓   | ✓   | ✓   | ✓   | —   | —   | —   |

**Table 12:** Coordination status with DG and the revised fuses: single-setting R2 against dual-setting R2.

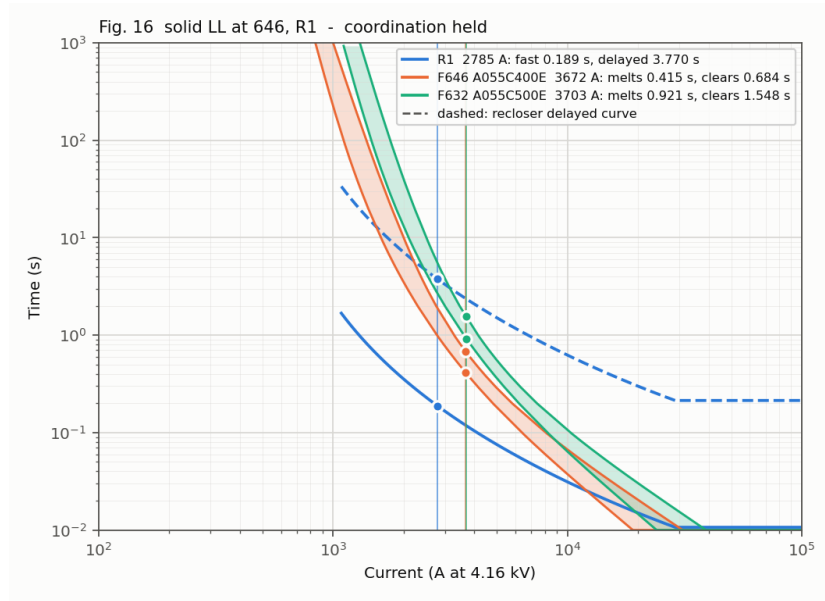


**Figure 11:** Coordination status with DG and the DSDR.

Figures 12 and 13 show a bolted LL fault at 646. The fuses carry 3.67 kA. R2 carries 1.11 kA in reverse and trips on its reverse fast curve in 0.086 s, before F646 melts (0.415 s).



**Figure 12:** Bolted LL fault at 646: R2 reverse setting with F646 and F632.



**Figure 13:** Bolted LL fault at 646: R1 with F646 and F632.

## 5.10 Operating Times With the DSDR (Table IV)

Tables 13 and 14 give the operating times with the DSDR. R1 operates forward for every node. R2 operates in reverse for the nodes upstream of it and forward for the nodes downstream. The fuse time is the minimum-melting time of the fuse nearest to the fault at the current through that

fuse. In all nine rows with a fuse it is longer than the last fast trip; the smallest margins are 6 ms at 675 and 15 ms at DL and 652.

|             |              | $I_{R1}$   | <b>R1 fast</b> | <b>R1 delayed</b> | $I_{R2}$   |             | <b>R2 fast</b> | <b>R2 delayed</b> |
|-------------|--------------|------------|----------------|-------------------|------------|-------------|----------------|-------------------|
| <b>Node</b> | <b>Fault</b> | <b>(A)</b> | <b>(s)</b>     | <b>(s)</b>        | <b>(A)</b> | <b>Dir.</b> | <b>(s)</b>     | <b>(s)</b>        |
| 632         | LLL          | 4734       | 0.081          | 1.627             | 1813       | rev         | 0.051          | 1.416             |
| 633         | LLL          | 3970       | 0.106          | 2.111             | 1500       | rev         | 0.060          | 2.001             |
| 645         | LL           | 3124       | 0.155          | 3.095             | 1232       | rev         | 0.075          | 3.002             |
| 646         | LL           | 2785       | 0.189          | 3.770             | 1114       | rev         | 0.086          | 3.771             |
| DL          | LLL          | 3954       | 0.106          | 2.123             | 1972       | rev         | 0.048          | 1.213             |
| 671         | LLL          | 3412       | 0.134          | 2.677             | 3391       | fwd         | 0.054          | 1.600             |
| 692         | LLL          | 3412       | 0.134          | 2.677             | 3391       | fwd         | 0.054          | 1.600             |
| 675         | LLL          | 3090       | 0.158          | 3.152             | 3050       | fwd         | 0.059          | 1.941             |
| 684         | LL           | 2543       | 0.222          | 4.437             | 2543       | fwd         | 0.072          | 2.813             |
| 680         | LLL          | 2789       | 0.188          | 3.761             | 2757       | fwd         | 0.066          | 2.381             |
| 611         | LG           | 1814       | 0.435          | 8.709             | 1718       | fwd         | 0.128          | 6.986             |
| 652         | LG           | 1795       | 0.446          | 8.919             | 1761       | fwd         | 0.123          | 6.570             |

**Table 13:** Recloser operating times with the DSDR, DG in service.

|             |              |             |             | $I_{fuse}$ | $t_{MMT}$  | $t_{TCT}$  | <b>Last fast</b> | <b>Margin</b> |               |
|-------------|--------------|-------------|-------------|------------|------------|------------|------------------|---------------|---------------|
| <b>Node</b> | <b>Fault</b> | <b>Fuse</b> | <b>Size</b> | <b>(A)</b> | <b>(s)</b> | <b>(s)</b> | <b>trip (s)</b>  | <b>(s)</b>    | <b>Status</b> |
| 633         | LLL          | F633        | 400E        | 5426       | 0.149      | 0.235      | 0.106            | +0.043        | held          |
| 645         | LL           | F632        | 500E        | 4191       | 0.613      | 0.970      | 0.155            | +0.458        | held          |
| 646         | LL           | F646        | 400E        | 3672       | 0.415      | 0.684      | 0.189            | +0.226        | held          |
| DL          | LLL          | F-DL        | 400E        | 5902       | 0.121      | 0.193      | 0.106            | +0.015        | held          |
| 692         | LLL          | F671-1      | 400E        | 3391       | 0.525      | 0.891      | 0.054            | +0.471        | held          |
| 675         | LLL          | F692-R      | 250E        | 5022       | 0.065      | 0.106      | 0.059            | +0.006        | held          |
| 684         | LL           | F671-2      | 300E        | 3963       | 0.148      | 0.230      | 0.072            | +0.076        | held          |
| 611         | LG           | F684        | 200E        | 2236       | 0.271      | 0.414      | 0.128            | +0.143        | held          |
| 652         | LG           | F652        | 150E        | 2239       | 0.138      | 0.193      | 0.123            | +0.015        | held          |

**Table 14:** Fuse times with the DSDR, DG in service. Margin =  $t_{MMT}$  minus the last fast trip (R2 for faults below R2; the later of R1 and R2 reverse above it). Nodes 632, 671 and 680 are on the main feeder and have no fuse.

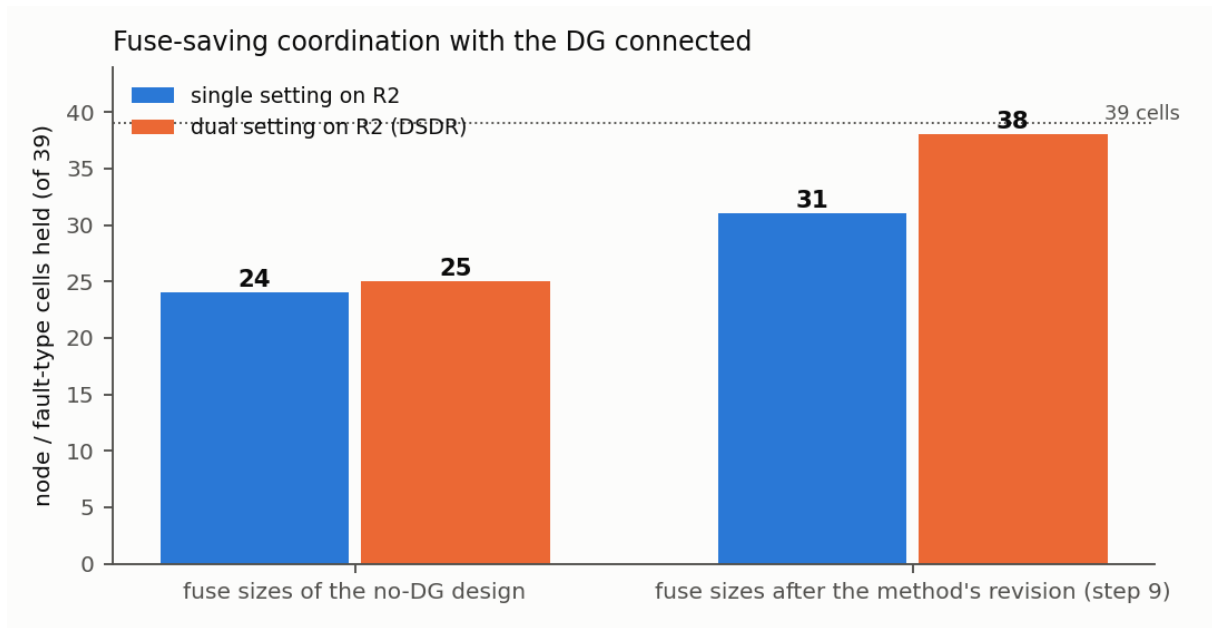
## 5.11 Single Setting Against Dual Setting on Selected Faults

Table 15 compares the two settings of R2 on nine faults with the same (revised) fuses: the six faults of the figures above and three cells of the classification. Figure 14 summarises it and Figures 15 and 16 show two of the diagrams.

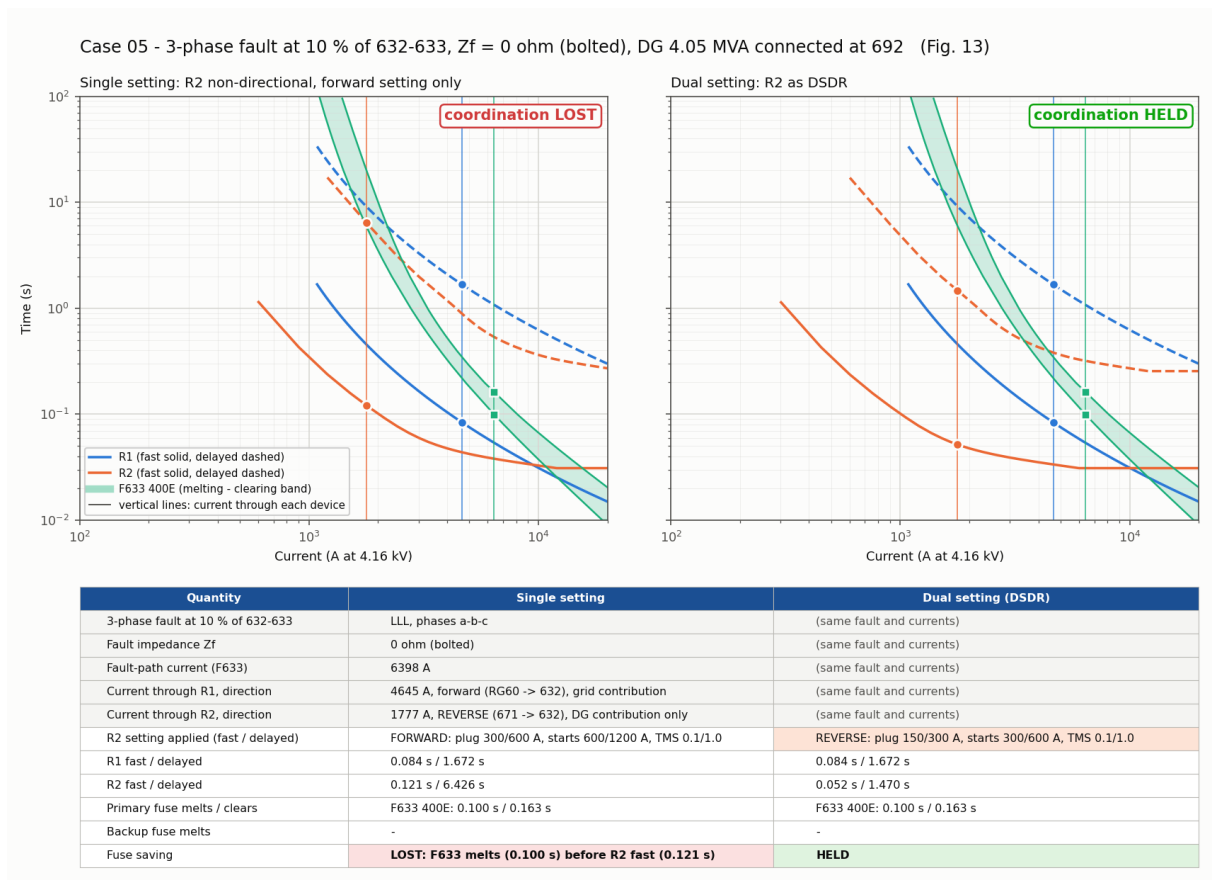
| #  | Figure            | Fault                                 | $Z_f$<br>( $\Omega$ ) | $I_{R1}$<br>(A) | $I_{R2}$ (A),<br>direction | R1 fast<br>(s) | R2 fast (s) |       | Fuse: melts<br>(s) | Fuse saving |      |
|----|-------------------|---------------------------------------|-----------------------|-----------------|----------------------------|----------------|-------------|-------|--------------------|-------------|------|
|    |                   |                                       |                       |                 |                            |                | single      | dual  |                    | single      | dual |
| 01 | Fig. 8            | LG fault at 611                       | 0                     | 1814            | 1718 (fwd)                 | 0.435          | 0.128       | 0.128 | F684 200E: 0.271   | held        | held |
| 02 | Fig. 9            | LLG fault at the middle of<br>692-675 | 1                     | 1445            | 1310 (fwd)                 | 0.742          | 0.204       | 0.204 | F692-R 250E: 0.618 | held        | held |
| 03 | Fig. 11           | LL fault at 646                       | 1                     | 1899            | 847 (rev)                  | 0.395          | 0.500       | 0.131 | F646 400E: 1.734   | lost        | lost |
| 04 | Fig. 12           | LL fault at 645                       | 1.5                   | 1694            | 783 (rev)                  | 0.507          | 0.605       | 0.149 | F632 500E: 9.129   | lost        | lost |
| 05 | Fig. 13           | 3-phase fault at 10 % of<br>632-633   | 0                     | 4645            | 1777 (rev)                 | 0.084          | 0.121       | 0.052 | F633 400E: 0.100   | lost        | held |
| 06 | Figs. 15, 16      | LL fault at 646                       | 0                     | 2785            | 1114 (rev)                 | 0.189          | 0.278       | 0.086 | F646 400E: 0.415   | held        | held |
| 07 | Fig. 14 / 17 cell | LG fault at 646 (phase c)             | 0                     | 2359            | 469 (rev)                  | 0.255          | no trip     | 0.396 | F646 400E: 1.057   | lost        | held |
| 08 | Fig. 14 / 17 cell | LG fault at 633 (phase b)             | 0                     | 2874            | 644 (rev)                  | 0.178          | 0.974       | 0.210 | F633 400E: 0.502   | lost        | held |
| 09 | Fig. 14 / 17 cell | LG fault at 692 (phase a)             | 0                     | 2197            | 2182 (fwd)                 | 0.293          | 0.089       | 0.089 | F671-1 400E: 2.842 | lost        | lost |

**Table 15:** Nine faults with the DG in service: single-setting against dual-setting R2, same fuses.

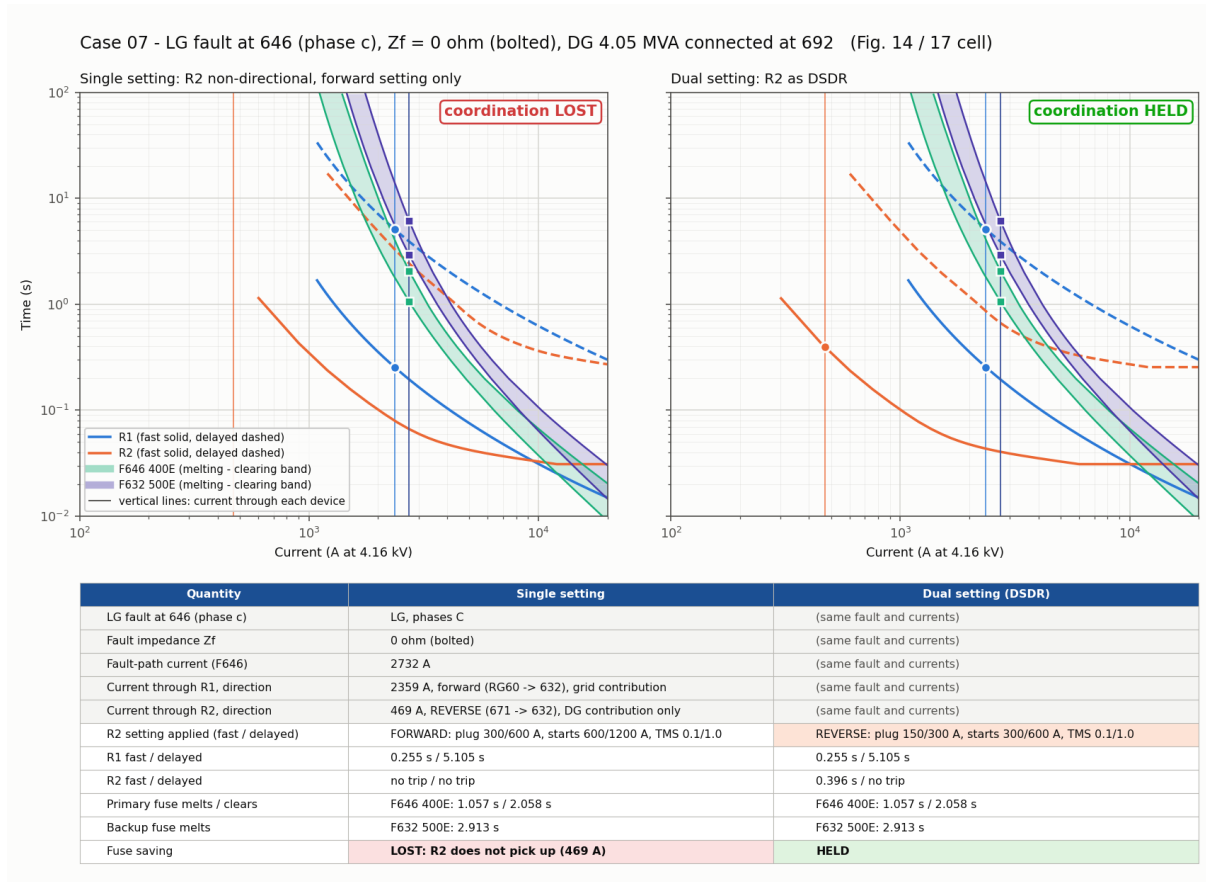
In cases 5, 7 and 8 the dual setting restores the fuse saving: the reverse group picks up a DG current that the forward group does not see (case 7: 469 A) or sees only slowly. Cases 3, 4 and 9 are lost with both settings; they are limits of the fuse revision and of the dial range, not of the direction. Cases 3 and 4 are faults through impedance: the larger fuses that are needed for the bolted faults are too slow for these lower currents.



**Figure 14:** Summary of the nine faults: single setting against dual setting.



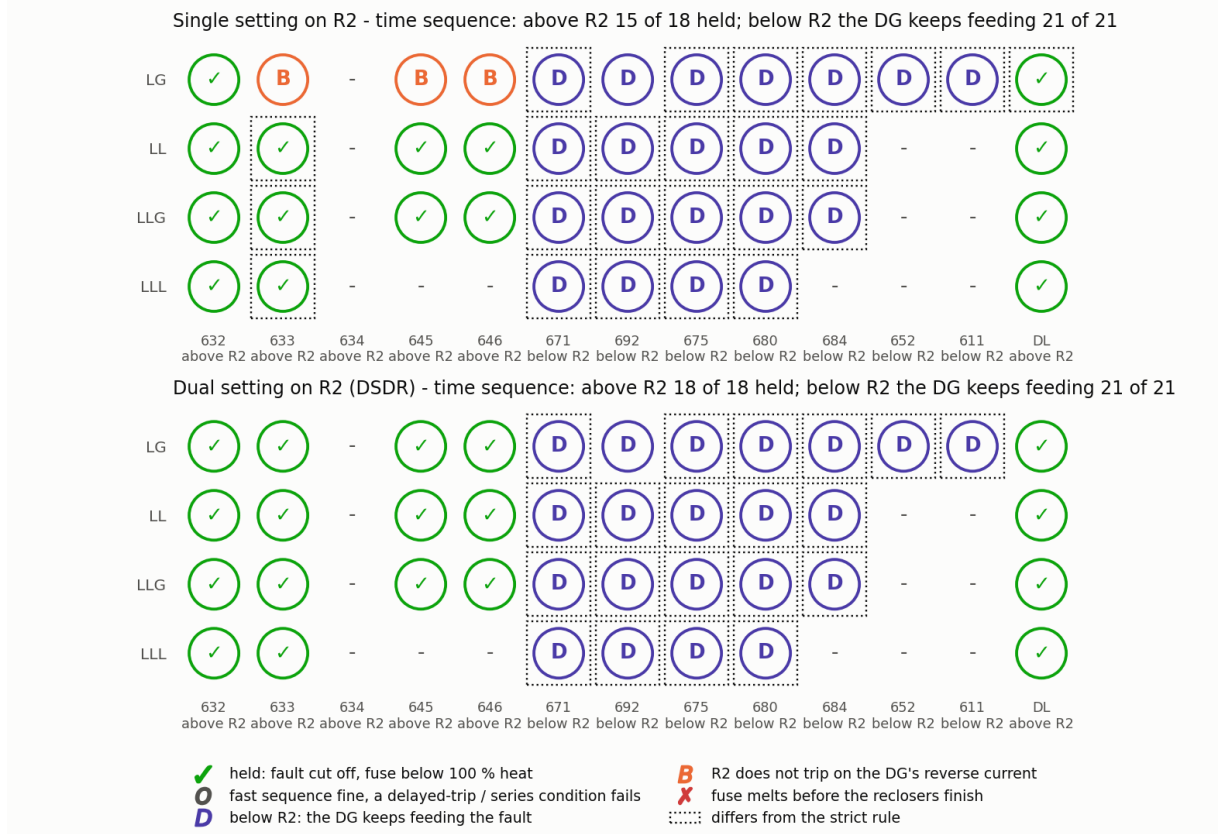
**Figure 15:** Three-phase fault at 10 % of line 632–633: single setting (left) and dual setting (right).



**Figure 16:** LG fault at 646: the single-setting R2 does not pick up the reverse current; the reverse group does.

## 5.12 Time-Sequence Check

The classification above uses the strict rule that every recloser must trip before the fuse starts to melt. A time-sequence check follows each fault in time instead: the fuse heats with the current that still flows after each recloser opens. For the 18 cells upstream of R2 the strict rule gives 11 cells held with the single setting and 18 with the dual setting; the time sequence gives 15 and 18 (Figure 17). The three cells that the DSDR restores in the time sequence as well are the LG faults at 633, 645 and 646, where the single-setting R2 never trips on a reverse current of about 0.5 to 0.6 kA. For the 21 cells downstream of R2 the DG at 692 feeds the fault directly, with no recloser in its path, and the result is the same for both settings.



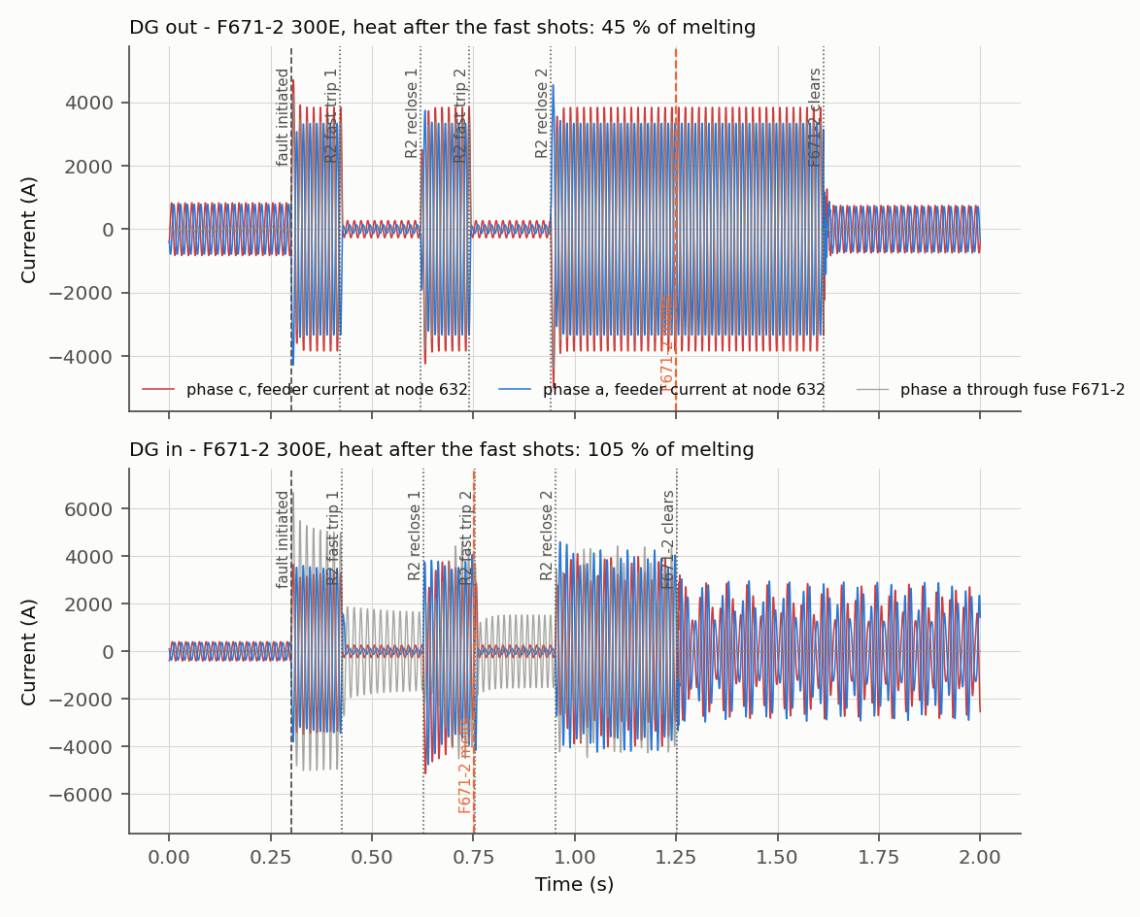
**Figure 17:** Time-sequence check of all 39 cells, single against dual setting.

### 5.13 Time-Domain Verification (Fig. 10)

Figure 18 is the EMT simulation of an LL (a–c) fault at 684 through  $0.2 \Omega$  with the conventional settings. The recloser trips are scripted from R2's fast curve applied to the simulated current, and the fuse F671-2 (300E) melts when the accumulated heating  $\sum \Delta t / t_{MMT}(I)$  reaches one. Without DG the two fast shots use 45 % of the melting heat, the fuse survives both and then clears the permanent fault. With the DG the fuse melts at the end of the second fast shot, because the DG keeps feeding the fault while R2 is open (Table 16). The feeder current at node 632 is 557 / 588 A RMS (phases a / c) before the fault and 2362 / 2719 A during it, with peaks of 3338 / 3842 A.



Fig. 10 - current at node 632, LL a-c fault at 684 through 0.2 ohm, PowerFactory EMT



**Figure 18:** Time-domain current at node 632 for an LL (a-c) fault at 684 through  $0.2 \Omega$  : without DG (top) and with DG (bottom).

| Event (time in s)              | DG out | DG in |
|--------------------------------|--------|-------|
| Fault initiated                | 0.30   | 0.30  |
| R2 fast trip 1                 | 0.420  | 0.426 |
| R2 reclose 1                   | 0.620  | 0.626 |
| R2 fast trip 2                 | 0.740  | 0.753 |
| R2 reclose 2                   | 0.940  | 0.953 |
| Fuse F671-2 melts              | 1.248  | 0.752 |
| Fuse operation / clears        | 1.613  | 1.253 |
| Fuse heat after the fast shots | 45 %   | 105 % |

**Table 16:** Reclosing sequence of the EMT simulation.

## 5.14 Verification With the Relay and Fuse Models of PowerFactory

The operating times of Chapter Five are calculated in Python from the library curves. As a check, the settings were written into the relay and fuse elements of the model, and PowerFactory itself calculated the tripping time of every device for every fault, for the conventional and for the DSDR settings. Over 1896 operating times the largest deviation is 1.22 %.

## 5.15 Sensitivity of the Result to the Margin

The 38 held cells are obtained with a zero margin. If the breaker interrupting time of three cycles is added to the fast trip, 31 cells are held; if in addition the fuse may only be loaded to 75 % of its melting time, 26 cells are held. The pickups are a second limit: with the DG, the lowest LG fault current through  $3\ \Omega$  is 577 A at R1 and 368 A at R2, below the pickups of 734.6 A and 587.5 A, because the DG supplies part of the fault.

## 5.16 Effect of the DG Penetration Level

To see how the conventional coordination degrades as the DG grows, seven separate PowerFactory projects were made, copies of the model with the DG at 692 set to 0, 10, 25, 37, 50, 75 and 100 % of 4.05 MVA. The machine, its step-up transformer and its active and reactive power are scaled together, so the per-unit data of Table 2 stay the same; at 0 % the DG is out of service. Every model keeps the conventional protection of the design without DG (single-setting R2, fuses of the design without DG), and all operating times are PowerFactory's own relay and fuse calculations. Two fuse–recloser pairs are followed: fuse F633 against R1 for a three-phase fault at 633 (R1's zone), and the lateral fuse F671-2 at node 671 against R2 for a line-to-line fault at 684 (R2's zone). The coordination time interval is

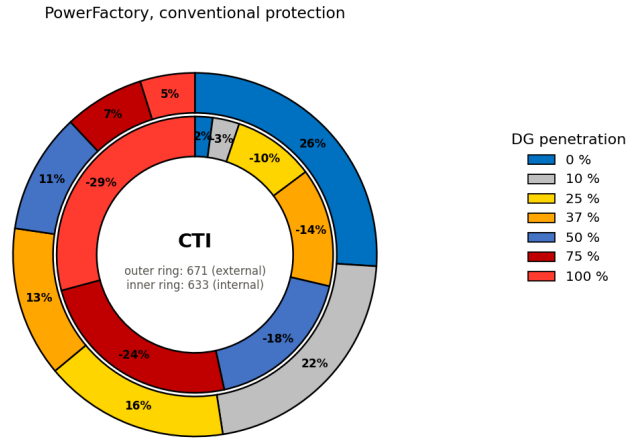
$$CTI = t_{MMT}(\text{fuse}) - t_F(\text{recloser}), \quad (19)$$

and, for the doughnut chart of Figure 19, each level's share of the ring,  $CTI\% = 100 \times CTI / \sum |CTI|$  over the seven levels, so that each ring adds up to 100 %.

| DG (%) | MVA  | Node 633: F633 and R1 (LLL) |                                       |          |        | Node 671: F671-2 and R2 (LL at 684) |                                       |          |        |
|--------|------|-----------------------------|---------------------------------------|----------|--------|-------------------------------------|---------------------------------------|----------|--------|
|        |      | I fuse / R1 (A)             | t <sub>MMT</sub> / t <sub>F</sub> (s) | CTI (ms) | ring % | I fuse / R2 (A)                     | t <sub>MMT</sub> / t <sub>F</sub> (s) | CTI (ms) | ring % |
| 0      | 0.00 | 4102 / 4162                 | 0.102 / 0.098                         | +4       | +2.1   | 2574 / 2711                         | 0.470 / 0.067                         | +403     | +26.0  |
| 10     | 0.41 | 4256 / 4142                 | 0.094 / 0.099                         | −5       | −3.1   | 2715 / 2681                         | 0.401 / 0.068                         | +333     | +21.5  |
| 25     | 1.01 | 4478 / 4111                 | 0.083 / 0.100                         | −17      | −9.6   | 2926 / 2636                         | 0.323 / 0.069                         | +254     | +16.4  |
| 37     | 1.50 | 4648 / 4088                 | 0.077 / 0.101                         | −24      | −13.9  | 3094 / 2598                         | 0.277 / 0.070                         | +207     | +13.4  |
| 50     | 2.02 | 4824 / 4062                 | 0.071 / 0.102                         | −31      | −17.9  | 3276 / 2557                         | 0.237 / 0.072                         | +166     | +10.7  |
| 75     | 3.04 | 5140 / 4015                 | 0.062 / 0.104                         | −42      | −24.2  | 3623 / 2511                         | 0.184 / 0.073                         | +110     | +7.1   |
| 100    | 4.05 | 5426 / 3970                 | 0.055 / 0.106                         | −51      | −29.2  | 3963 / 2543                         | 0.147 / 0.072                         | +75      | +4.9   |

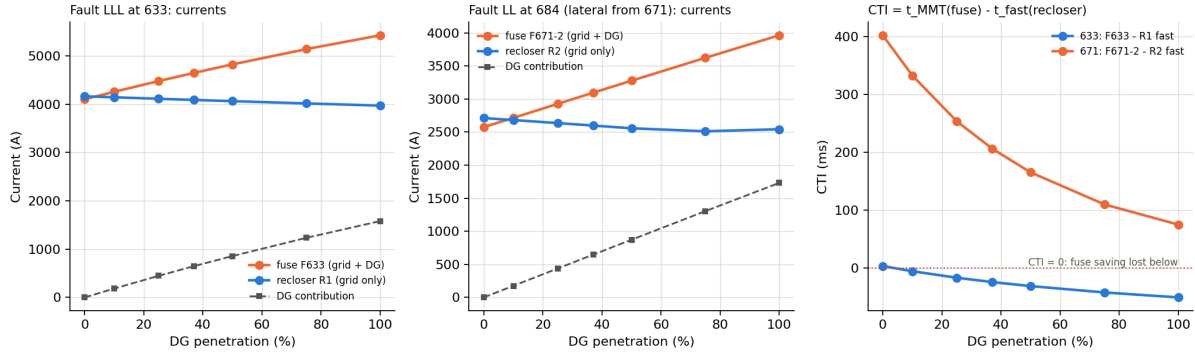
**Table 17:** CTI against DG penetration, conventional protection.

Fig. 7 - percentage variation of the CTI with DG penetration (each ring = 100 %; negative = fuse melts b



**Figure 19:** Percentage variation of the CTI with DG penetration (each ring adds up to 100 %; outer ring 671, inner ring 633).

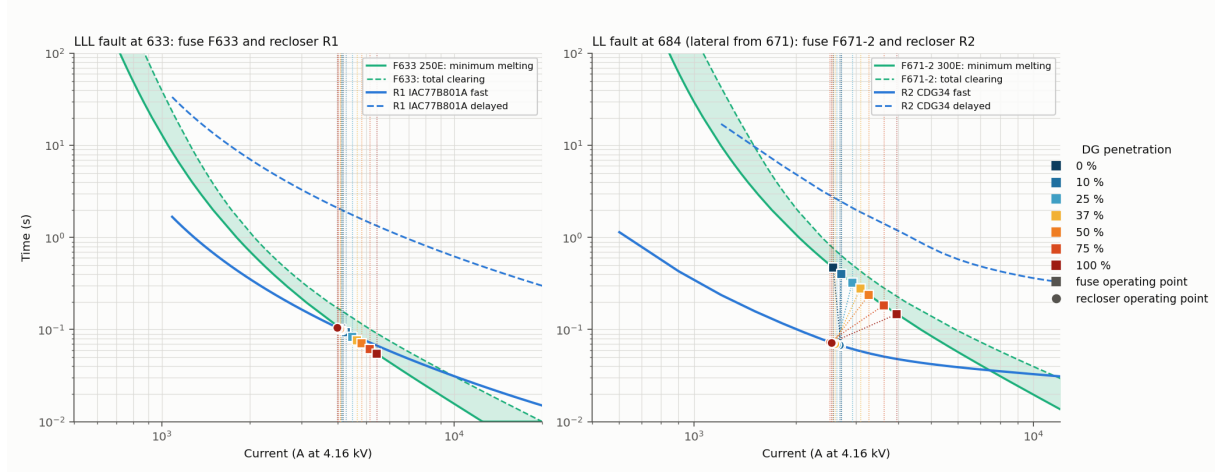
At node 633 the fuse current rises from 4102 A to 5426 A while R1's current falls from 4162 A to 3970 A; the CTI falls from +4 ms to −51 ms. At node 671 the fuse current rises from 2574 A to 3963 A while R2's current stays at about 2543 A; the CTI falls from +403 ms to +75 ms. The mechanism is the same at both nodes (Figure 20): the fuse lies between the fault and both sources and carries the grid share and the DG share, while the recloser carries only the grid share, which even falls slightly because the DG holds up the voltage of the network during the fault. A fuse curve is steep, so the extra current shortens its melting time much more than the lower current lengthens the recloser's time, and the CTI decreases. At 633 the margin is only a few milliseconds without DG and is lost from about 4 % penetration; at 671 the margin is large and stays positive up to 100 %.



**Figure 20:** Currents through the fuse (grid and DG share), the recloser (grid share) and from the DG, and the resulting CTI, against DG penetration.

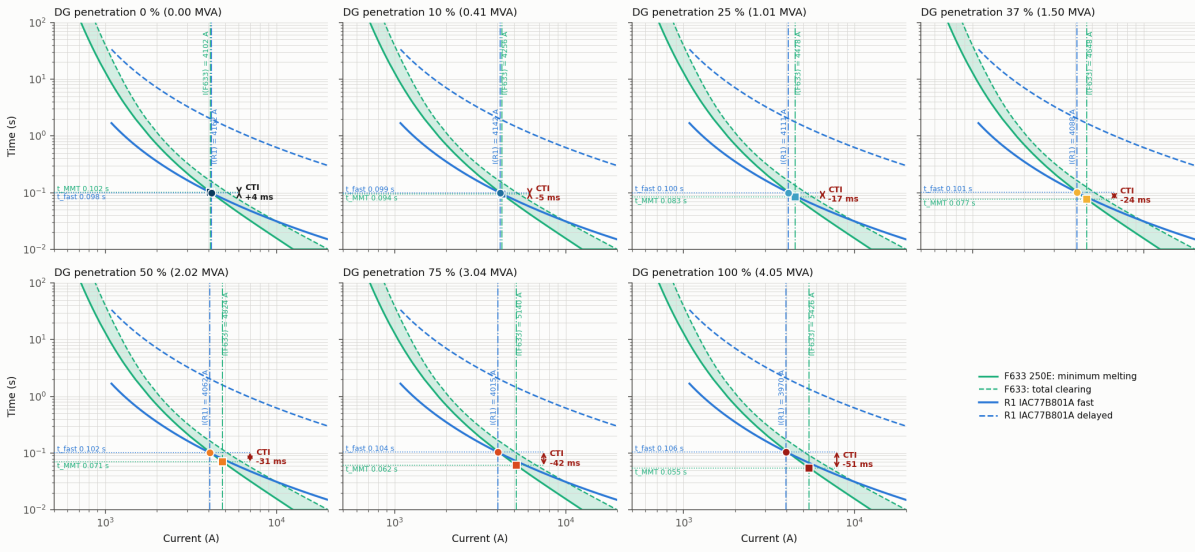
Figure 21 shows the same on the time–current characteristics: the curves are fixed because the settings are not changed, the recloser’s operating point stays where it is, and the fuse’s operating point moves to a higher current and a shorter time with every step. Figure 22 gives node 633 level by level, with the vertical short-circuit lines of the fuse and the recloser currents, the operating times and the CTI.

TCC with increasing DG penetration (conventional settings): the recloser point stays, the fuse point moves right and down



**Figure 21:** Time–current characteristics at all penetration levels: fuse operating points (squares) and recloser operating points (circles).

TCC - LLL fault at 633: fuse F633 and recloser R1, conventional settings, at each DG penetration level (square: fuse, circle: recloser)



**Figure 22:** Node 633, fuse F633 and R1, at each penetration level, with the short-circuit currents, operating times and CTI.

The fault level rises at every node (Table 18), most near the DG (671, 692, 675: about +60 %) and least behind the 4.16/0.48 kV transformer (634) and on the single-phase laterals. The current through the reclosers does not rise with it.

| Node | Fault   | 0 %   | 25 %  | 50 %  | 75 %  | 100 % | Change |
|------|---------|-------|-------|-------|-------|-------|--------|
| 632  | LLL ABC | 4735  | 5227  | 5690  | 6123  | 6526  | +38 %  |
| 633  | LLL ABC | 4102  | 4478  | 4824  | 5140  | 5426  | +32 %  |
| 634  | LLL ABC | 15671 | 16307 | 16828 | 17257 | 17608 | +12 %  |
| 645  | LLG BC  | 3418  | 3685  | 3932  | 4149  | 4341  | +27 %  |
| 646  | LLG BC  | 3112  | 3332  | 3528  | 3701  | 3854  | +24 %  |
| 671  | LLL ABC | 3390  | 3919  | 4461  | 5010  | 5564  | +64 %  |
| 692  | LLL ABC | 3390  | 3919  | 4461  | 5010  | 5564  | +64 %  |
| 675  | LLL ABC | 3151  | 3612  | 4087  | 4557  | 5022  | +59 %  |
| 680  | LLL ABC | 2930  | 3322  | 3711  | 4091  | 4460  | +52 %  |
| 684  | LLG AC  | 2655  | 3027  | 3395  | 3756  | 4107  | +55 %  |
| 652  | LG A    | 1874  | 1983  | 2080  | 2164  | 2239  | +19 %  |
| 611  | LG C    | 1857  | 1973  | 2074  | 2161  | 2236  | +20 %  |
| DL   | LLL ABC | 3946  | 4454  | 4952  | 5436  | 5902  | +50 %  |

**Table 18:** Largest bolted fault current at each node (A) against DG penetration.

The load flow changes as well (Table 19). The DG supplies part of the load locally, so the load current through both reclosers falls, the voltages rise, and the power through R2 reverses between 75 and 100 %. By eq. (3) the pickups would have to fall with the penetration; with the fixed settings (R1 720 A, its curve starting at 1080 A; R2 fast curve starting at 600 A) the reclosers no longer detect the weakest faults (LG through  $3\ \Omega$ ) in their zones, R2 from just above 50 %.

| DG (%) | DG P (MW) | R1 load (A) | R2 load (A) | R2 flow | $V_{671,a}$ (p.u.) | R1 $1.25 I_{nom}$ (A) | R2 $1.25 I_{nom}$ (A) | R1 lowest $3\ \Omega$ fault (A) | R2 lowest $3\ \Omega$ fault (A) |
|--------|-----------|-------------|-------------|---------|--------------------|-----------------------|-----------------------|---------------------------------|---------------------------------|
| 0      | 0.00      | 587.7       | 470.0       | forward | 0.988              | 734.6                 | 587.5                 | 1074*                           | 878                             |
| 10     | 0.32      | 543.1       | 430.1       | forward | 0.994              | 678.9                 | 537.6                 | 1018*                           | 821                             |
| 25     | 0.81      | 480.7       | 372.2       | forward | 1.002              | 600.9                 | 465.2                 | 937*                            | 738                             |
| 37     | 1.20      | 434.8       | 328.6       | forward | 1.009              | 543.5                 | 410.8                 | 874*                            | 673                             |
| 50     | 1.62      | 389.9       | 285.3       | forward | 1.015              | 487.4                 | 356.6                 | 809*                            | 606                             |
| 75     | 2.43      | 321.2       | 222.4       | forward | 1.025              | 401.5                 | 278.0                 | 689*                            | 483*                            |
| 100    | 3.24      | 284.9       | 252.2       | reverse | 1.033              | 356.1                 | 315.2                 | 577*                            | 368*                            |

**Table 19:** Load flow, pickups by eq. (3) and lowest fault current through each recloser against DG penetration. \*: below the start of the recloser's curve with the fixed settings (not detected).

The penetration study therefore shows the two ways in which fixed settings fail as the DG grows: the fuse saving is lost because the fuse speeds up while the recloser does not, and the sensitivity is lost because the currents through the reclosers fall. Both are reasons for settings that adapt to the DG, which is what the dual-setting directional recloser provides.

## 5.17 Results Summary

| Stage                                                     | Cells held (of 39) |
|-----------------------------------------------------------|--------------------|
| A. No DG, starting fuse sizes                             | 35                 |
| A. No DG, after the fuse revision (F692-R to 200E)        | 39                 |
| B. DG connected, conventional R2 (Fig. 14)                | 24                 |
| B. DG connected, no-DG fuses, R2 as DSDR                  | 25                 |
| C. DG connected, revised fuses, conventional R2           | 31                 |
| C. DG connected, revised fuses, R2 as DSDR (Fig. 17)      | 38                 |
| C. The same settings and fuses with the DG out of service | 37                 |

**Table 20:** Summary of the coordination results.

The dual setting alone, with the fuses of the design without DG, restores one cell (24 to 25). Together with the fuse revision it restores fourteen (24 to 38), seven of which need the dual setting.

## **CHAPTER SIX: DISCUSSION**

### **6.1 Effect of DG on Fault Current**

The DG raises the fault current at every node, most near its connection point (671 and 692: from 3.39 to 5.56 kA). The fuses carry this higher current and melt faster, while R1 carries about the same grid current as before. The margin between the fuse and the fast trip, which is small even without DG, disappears.

### **6.2 Effect of Fault Current Direction**

For faults at 632, 633, 645, 646 and DL the current through R2 is reverse and consists only of the DG contribution. Its magnitude (0.5 to 2.0 kA) is much lower than the forward fault currents for which the forward group is set. A forward group with plugs of 300 and 600 A responds slowly to these currents, or not at all for LG faults.

### **6.3 Performance of the Conventional Protection**

With the conventional single setting the scheme keeps coordination at the nodes on the main feeder and loses it on the fused laterals of R1's zone and at 675. The loss depends on the location and on the fault type; it is not a uniform failure of the feeder.

### **6.4 Performance of the DSDR**

The reverse group, with half the plug settings, trips for every reverse fault before the fuse melts once the fuses are revised. The result of 38 cells against 31 with the same fuses isolates its contribution. Without the fuse revision the improvement is one cell only: at 633 and DL the fuses of the design without DG melt in 0.05 to 0.16 s with the DG, before R1's fast trip (0.11 to 0.18 s), and no setting of R2 can change R1's time. The DSDR is therefore necessary but not sufficient.

### **6.5 Limits Reached by the Devices**

R1's dials are the two extremes of the IAC range, and the CDG table ends at 0.1 and 1.0. The method's dial revision has no room. The fuse revision creates a conflict between the two states



of the network: the larger fuses that are needed with the DG are too slow for two cells without it, and for the faults through impedance. Fuse sizes cannot adapt to the DG state. For faults downstream of R2 the fuse saving relies on the DG's own protection disconnecting it during the dead time, which is not part of the method.

## **6.6 Change of the Simulation Tool**

Rebuilding the study in PowerFactory removed the main limitations of the earlier MATLAB/Simulink phase. The DG is a dynamic synchronous machine with its full data instead of a static source. The reclosers use the curves of the actual devices instead of a substitute curve. All 39 cells are resolved, including the three-phase faults and the single-phase nodes that could not be built before. The complete DSDR stage, which was pending, is finished. The earlier model remains useful as an independent cross-check of the load flow.

## **6.7 Engineering Implication**

Adaptive directional protection is justified where a DG is downstream of a mid-line recloser. To apply it, the engineer must check the dial range of the actual relay, revise the fuse sizes for the DG state, verify the pickup against the reduced minimum fault current with DG, and provide for the DG infeed to faults beyond the recloser.

# CHAPTER SEVEN: CONCLUSION AND FUTURE WORK

## 7.1 Conclusion

The DSDR method of Yousaf *et al.* was implemented on the IEEE 13-node feeder in DIgSILENT PowerFactory with the real characteristics of the protection devices. The feeder model agrees with the IEEE short-circuit benchmark within about 2 %. Without DG the conventional scheme is coordinated in all 39 cells. With the 4.05 MVA DG at node 692 and a conventional R2, coordination is held in 24 cells. With R2 as a DSDR and the fuse revision of the method it is held in 38 cells; with the same fuses and a single setting in 31. The dual-setting directional recloser therefore restores the coordination in the zone between the grid and the DG, with the qualification that the result depends on the fuse revision and that one cell and the faults downstream of R2 remain open.

## 7.2 Major Findings

1. Fault levels at the IEEE 13-node benchmark (feeder head 4.73 kA).
2. R1: pickup 720 A, TDS 0.5 / 10, GE IAC extremely inverse curve.
3. R2 forward: plugs 300 / 600 A, TMS 0.1 / 1.0; R2 reverse: plugs 150 / 300 A, TMS 0.1 / 1.0; reverse pickup by eq. (12) 315.3 A.
4. With DG the load current at R2 reverses (252 A) and R2 carries 1.1 to 2.0 kA in reverse for upstream faults.
5. Coordination: 39 cells without DG; 24 with DG and a conventional R2; 38 with the DSDR.
6. The dual setting restores seven cells with the revised fuses and one cell without the revision.
7. The LG fault at 692 cannot be coordinated within the dial range of R2.
8. With rising DG penetration and fixed settings the CTI falls at both studied nodes (633: from about +4 to -51 ms; 671: from +403 to +75 ms), the fault level rises by up to 64 %, and R2 stops detecting the weakest fault of its zone above about 50 %.
9. PowerFactory's own relay and fuse models confirm the operating times within 1.22 %.

10. In the EMT simulation the fuse survives the two fast shots without DG and melts at 1.25 s; with DG it melts during the second fast shot.

### **7.3 Contributions of the Present Work**

- A reproducible PowerFactory implementation of the method: one command rebuilds the model and all results.
- Separation of the effect of the dial setting from the effect of the fuse revision.
- Identification of the limits of the method: dial range, DG-in against DG-out fuse sizes, DG infeed downstream of the recloser, and sensitivity of the pickups with DG.
- A time-domain verification and a verification with the relay models of the tool.

### **7.4 Remaining Work**

1. Extend the study to the IEEE 34-node feeder.
2. Model the directional element of R2 explicitly instead of assigning the direction from the fault location.
3. Repeat the classification with a practical margin (breaker time and 75 % of the fuse melting time).

### **7.5 Future Work**

Future work may include a recloser with a wider dial range or with user-defined curves, a setting-group change triggered by the DG state, the disconnection logic of the DG during the reclosing dead time, the sensitivity to fault impedance, inverter-based generation, and hardware-in-the-loop tests.

## REFERENCES

- [1] M. Yousaf, A. Jalilian, K. M. Muttaqi, and D. Sutanto, “An adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation,” *IEEE Transactions on Industry Applications*, vol. 58, no. 2, pp. 1831–1842, Mar./Apr. 2022, doi: 10.1109/TIA.2022.3146095.
- [2] W. H. Kersting, “Radial distribution test feeders,” *IEEE Transactions on Power Systems*, vol. 6, no. 3, pp. 975–985, Aug. 1991.
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- [4] A. Naiem, Y. Hegazy, A. Abdelaziz, and M. Elsharkawy, “A classification technique for recloser-fuse coordination in distribution systems with distributed generation,” *IEEE Transactions on Power Delivery*, vol. 27, no. 1, pp. 176–185, Jan. 2012.
- [5] M. Yousaf, K. M. Muttaqi, and D. Sutanto, “Overcurrent protection scheme for the IEEE 13-node benchmark test feeder with improved selectivity,” in *Proc. IEEE Power & Energy Society General Meeting*, 2020, doi: 10.1109/PESGM41954.2020.9281529.
- [6] W. H. Kersting and G. Shirek, “Short circuit analysis of IEEE test feeders,” in *Proc. IEEE PES Transmission and Distribution Conference and Exposition*, 2012.
- [7] DIgSILENT GmbH, *PowerFactory 2021 User Manual*, Gomaringen, Germany, 2021.

## **APPENDIX A: COMPLETE COPY OF THE REFERENCE PAPER**

The complete reference paper by Yousaf *et al.* [1] is attached as a separate, unchanged PDF copy in accordance with the requirement of the department. It is not reproduced inside this document.

## APPENDIX B: POWERFACTORY MODEL AND PARAMETER NOTES

- Project: *IEEE13 DSDR Fuse Coordination*, imported from `IEEE 13 Node Feeder.pfd`; study case *Study with Substation Transformer*.
- Source: external grid 115 kV, substation transformer 115/4.16 kV, 5 MVA,  $u_k = 8.06\%$ .
- Regulator taps fixed at +10 (a), +8 (b), +11 (c).
- DG: synchronous machine 4.05 MVA, 0.69 kV, round rotor, data of Table 2; transformer 4.16/0.69 kV,  $u_k = 15\%$ ; operating point 3.24 MW, voltage control at 1.0 p.u.
- R1: relay type IAC77B801A at RG60–632, units “R1 Fast” and “R1 Delayed”.
- R2: relay type CDG34 at the 671 end of line 632–671, units “R2 Fast”, “R2 Delayed”, “R2 Rev Fast” and “R2 Rev Delayed”.
- Fuses: fuse elements with the library types A055C (4.16 kV) and gL-800A (0.48 kV).
- The protection devices are created in the base network. Changes made while the substation-transformer variation is active are recorded in that variation and are removed by the scripts after each run.
- PowerFactory must be closed when the scripts are run from Python; inside PowerFactory the script object `Final(1)` runs the same sequence.

## APPENDIX C: IMPLEMENTATION FORMULAS

The operating times are evaluated with the following expressions, where  $M = I_f/I_p$  for R1 and  $M = I_f/I_s$  for R2.

$$\text{R1 (IAC): } t = TDS \left[ 0.004 + \frac{0.6379}{M - 0.62} + \frac{1.7872}{(M - 0.62)^2} + \frac{0.2461}{(M - 0.62)^3} \right] \quad (\text{A})$$

$$\text{R2 (CDG table): } t = t_1 \left( \frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(M/M_1)}{\ln(M_2/M_1)} \quad (\text{B})$$

$$\text{Fuse (curve points): } t = t_1 \left( \frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(I/I_1)}{\ln(I_2/I_1)} \quad (\text{C})$$

$$\text{Fuse saving: } t_{MMT}(I_{fuse}) > t_F, \quad t_{TCT}(I_{fuse}) < t_D \quad (\text{D})$$

$$\text{Series fuses: } t_{TCT,i} \leq 0.75 t_{MMT,i+1} \quad (\text{E})$$

$$\text{Fuse heating (EMT): } \sum_j \frac{\Delta t_j}{t_{MMT}(I_j)} = 1 \Rightarrow \text{the fuse melts} \quad (\text{F})$$

**Worked example, node 633, three-phase fault, DG in service.** R1 carries 3970 A:  $M = 3970/720 = 5.514$ ,  $M - 0.62 = 4.894$ , bracket  $= 0.004 + 0.1303 + 0.0746 + 0.0021 = 0.2111$ , so  $t_F = 0.5 \times 0.2111 = 0.106$  s and  $t_D = 10 \times 0.2111 = 2.111$  s. R2 carries 1500 A in reverse: fast,  $M = 1500/150 = 10.0$ , table 0.060 s; delayed,  $M = 1500/300 = 5.0$ , table 2.0 s. The fuse F633 (400E) carries 5426 A, the sum of both contributions: its curve points are 4800 A at 0.2 s and 6392 A at 0.1 s,  $f = \ln(5426/4800)/\ln(6392/4800) = 0.428$ ,  $t_{MMT} = 0.2 \times 0.5^{0.428} = 0.149$  s. The fuse outlasts the last fast trip (0.106 s) by 0.043 s.

## APPENDIX D: CURRENT PROJECT STATUS

| Work package                                      | Status    | Evidence                                                      |
|---------------------------------------------------|-----------|---------------------------------------------------------------|
| IEEE 13-node model in PowerFactory                | Completed | Rated currents within 2 %, fault levels at the IEEE benchmark |
| Unbalanced load flow                              | Completed | Tables 6 and 7                                                |
| Short-circuit analysis                            | Completed | All fault types and phase combinations, DG out and in         |
| DG connection (dynamic machine)                   | Completed | Table 2                                                       |
| Conventional coordination without DG              | Completed | 39 of 39 cells                                                |
| Classification with DG (Fig. 14)                  | Completed | 24 of 39 cells held                                           |
| DG penetration study                              | Completed | Section 5.16, seven PowerFactory models                       |
| DSDR forward and reverse settings                 | Completed | Table 3                                                       |
| DSDR coordination restoration (Fig. 17, Table IV) | Completed | 38 of 39 cells                                                |
| Time-domain verification (Fig. 10)                | Completed | Table 16                                                      |
| Verification with PowerFactory relay models       | Completed | 1896 operating times, max 1.22 %                              |
| IEEE 34-node extension                            | Pending   | Not started                                                   |

**Table 21:** Status of the work packages.



## **APPENDIX E: COMPARISON WITH THE REFERENCE PAPER**

This appendix compares the results of Chapter Five with the values printed in the reference paper [1] or read from its figures.

### **Basis of Comparison**

The comparison uses only values that the paper prints or that can be read from its figures. The model was not tuned to match the paper; differences are stated and explained from the evidence in the model.

| Aspect                                       | This project                              | Reference paper               | Assessment                       |
|----------------------------------------------|-------------------------------------------|-------------------------------|----------------------------------|
| Rated branch currents                        | within 2 %                                | Table II                      | Very good                        |
| Fault levels                                 | IEEE benchmark, feeder head 4.73 kA       | 5.41 kA, some branches higher | Lower                            |
| R1 worked example (4219 A)                   | 0.096 / 1.926 s                           | 0.097 / 1.932 s               | Very good                        |
| Recloser curves                              | IAC77B801A and CDG34 library curves       | Same devices                  | Same                             |
| Fuse coefficients, series fuses of R1's zone | within 0.05 with reversed index           | Table III                     | Index reversed in the paper      |
| DG model                                     | Dynamic synchronous machine, Table I data | Synchronous machine           | Same                             |
| Coordination with DG, conventional R2        | 24 of 39 held; 29 cells agree             | 30 of 39 held (Fig. 14)       | Good                             |
| Coordination with the DSDR                   | 38 of 39 held                             | 39 of 39 held (Fig. 17)       | Good; one cell at the dial limit |
| Fuse current, LL at 646                      | 3.67 kA                                   | 3.64 kA (Fig. 15)             | Very good                        |
| R2 reverse current, LL at 646                | 1.11 kA                                   | 1.54 kA                       | Lower                            |
| Table IV, R1 times                           | 12 to 78 % slower (652: 34 % faster)      | Table IV                      | Follows the fault levels         |
| Table IV, R2 and fuse times                  | assumed R2 settings, revised fuses        | settings not published        | Not comparable in detail         |
| EMT sequence, fuse operation                 | melts 1.25 s, clears 1.61 s               | 1.21 s (Fig. 10)              | Close if 1.21 s is melting       |
| IEEE 34-node feeder                          | not modelled                              | validated                     | Remaining work                   |

**Table 22:** Comparison of the results with the reference paper.

## Base-Case Comparison

The rated currents test the feeder model independently of the protection and of the DG. Their agreement within 2 % (the MATLAB/Simulink model of the earlier phase reached 2 to 9 %) shows that the topology, the loads and the regulator are implemented as in the paper.

## **Fault-Current Magnitude Difference**

The maximum fault currents of the feeder branches are 4 to 60 % below the paper's Table II. The model was validated against the published IEEE short-circuit results instead, which it meets within about 2 %. The paper's values cannot all be correct: 6.73 kA on 632–633 and 7.83 kA on 692–675 exceed its own 5.41 kA at the feeder head. The two worked cases of the paper illustrate the remaining difference: R2 carries 1718 A for the LG fault at 611 (paper: 1974 A) and 1114 A in reverse for the LL fault at 646 (paper: 1542 A). In the earlier MATLAB/Simulink phase these currents were 702 A and 838 A, because the DG was a static Thevenin source.

## **Fuse and Coordination Comparison**

The paper's Table III could be reproduced only with the series index reversed. The fuse times in the paper's figures do not follow its fuse equation: for F646 at 3.6 kA the figure shows about 0.18 s, the equation with Table III gives 1.85 s. The figures match A055C library fuses, and the sizes change between figures (F646 is 200E in Fig. 12 and 300E in Figs. 11 and 16). The model therefore uses library fuse curves throughout.

## **Operating-Time Comparison (Table IV)**

The R1 equation and settings reproduce the paper's worked example, so the difference of the R1 times comes from the fault currents. In every row of the paper the delayed time of R1 is 20 times the fast time, the ratio of the dials 10 and 0.5, which confirms the dials. For R2 the paper prints no settings and the ratio of its two times is not constant, so the R2 times and the fuse times of the model are the result of the stated method and cannot be matched to the paper row by row.

## **Central Finding Comparison**

The central finding of the paper is reproduced: the DG makes a conventional scheme lose coordination in the zone between the grid and the DG, and a dual-setting directional R2 restores it there. The model adds two qualifications. The dual setting needs the fuse revision to be effective; and it does not help for faults downstream of R2, which the DG feeds directly.

## Tables of the comparison

| Parameter                                      | Symbol     | Paper  | Model  | Unit |
|------------------------------------------------|------------|--------|--------|------|
| Leakage reactance                              | $X_l$      | 0.05   | 0.05   | pu   |
| Stator resistance                              | $R_a$      | 0.0014 | 0.0014 | pu   |
| d-axis synchronous reactance                   | $X_d$      | 1.4    | 1.4    | pu   |
| d-axis transient reactance                     | $X'_d$     | 0.231  | 0.231  | pu   |
| d-axis subtransient reactance                  | $X''_d$    | 0.118  | 0.118  | pu   |
| d-axis transient open-circuit time constant    | $T'_{d0}$  | 5.5    | 5.5    | s    |
| d-axis subtransient open-circuit time constant | $T''_{d0}$ | 0.05   | 0.05   | s    |
| q-axis synchronous reactance                   | $X_q$      | 1.372  | 1.372  | pu   |
| q-axis transient reactance                     | $X'_q$     | 0.8    | 0.8    | pu   |
| q-axis subtransient reactance                  | $X''_q$    | 0.118  | 0.118  | pu   |
| q-axis transient open-circuit time constant    | $T'_{q0}$  | 1.25   | 1.25   | s    |
| q-axis subtransient open-circuit time constant | $T''_{q0}$ | 0.19   | 0.19   | s    |
| Mechanical starting time                       | $M = 2H$   | 1.5    | 1.5    | s    |
| DG rating                                      | $S_n$      | 4.05   | 4.05   | MVA  |
| DG voltage                                     | $U_n$      | 0.69   | 0.69   | kV   |
| Step-up transformer, HV side                   | $U_{HV}$   | 4.16   | 4.16   | kV   |
| Step-up transformer leakage reactance          | $x_T$      | 0.15   | 0.15   | pu   |

**Table 23:** DG parameters: reference paper (Table I) against the PowerFactory model.

| Branch       | $I_{nom}$ (A) |       |        | $I_{f,min}$ (kA) |       | $I_{f,max}$ (kA) |       |        |
|--------------|---------------|-------|--------|------------------|-------|------------------|-------|--------|
|              | model         | paper | dev. % | model            | paper | model            | paper | dev. % |
| RG60–632     | 587.7         | 587.1 | +0.1   | 1.07             | 1.24  | 4.73             | 5.41  | −13    |
| 632–633      | 81.0          | 81.1  | −0.1   | 0.76             | 0.72  | 4.1              | 6.73  | −39    |
| 632–645      | 143.1         | 143.3 | −0.1   | 0.73             | 0.93  | 3.42             | 4.25  | −20    |
| 632–671      | 470.0         | 478.2 | −1.7   | 0.88             | 1.08  | 3.39             | 4.52  | −25    |
| XFM1-HV side | 81.0          | 81.1  | −0.1   | 0.08             | 0.73  | 1.81             | 6.48  | −72    |
| XFM1-LV side | 706.4         | 704.7 | +0.2   | 0.71             | 0.59  | 15.67            | 18.76 | −16    |
| 645–646      | 66.1          | 64.9  | +1.8   | 0.73             | 0.84  | 3.11             | 3.64  | −15    |
| 671–692      | 229.7         | 229.3 | +0.2   | 0.74             | 0.78  | 3.39             | 3.69  | −8     |
| 671–684      | 71.1          | 70.9  | +0.3   | 0.66             | 0.69  | 2.66             | 3.47  | −23    |
| 671–680      | 0.0           | 0.0   | –      | 0.64             | 0.59  | 2.93             | 6.25  | −53    |
| 692–675      | 205.9         | 205.9 | +0.0   | 0.74             | 0.71  | 3.15             | 7.83  | −60    |
| 684–611      | 71.1          | 70.8  | +0.4   | 0.68             | 0.63  | 1.86             | 1.93  | −4     |
| 684–652      | 63.0          | 63.1  | −0.2   | 0.66             | 0.66  | 1.87             | 2.67  | −30    |

**Table 24:** Branch currents: model against Table II of the paper. XFM1 HV side: the model values are 4.16 kV currents for a fault on the 0.48 kV side, so this row is not comparable.

| Fuse   | $I_f$ (A) | $i/z$ | $t_F$ (s) | $t_D$ (s) | $t_{fuse}$ (s) | $b_i$ | $b_i$ rev. | paper | Size    | $t_{MMT}$ (s) |
|--------|-----------|-------|-----------|-----------|----------------|-------|------------|-------|---------|---------------|
| F632   | 3418      | 2/2   | 0.133     | 2.670     | 1.824          | 6.62  | 6.35       | 6.38  | 400E    | 0.512         |
| F633   | 4102      | 1/1   | 0.100     | 2.008     | 1.054          | 6.53  | 6.53       | 6.83  | 250E    | 0.102         |
| F634   | 15671     | 1/2   | 0.439     | 8.772     | 3.217          | 8.06  | 8.33       | 8.32  | gL-800A | 0.021         |
| F645   | 3418      | 1/2   | 0.133     | 2.670     | 0.979          | 6.35  | 6.62       | 6.65  | 250E    | 0.158         |
| F646   | 3112      | 1/2   | 0.156     | 3.115     | 1.142          | 6.35  | 6.62       | 6.67  | 300E    | 0.273         |
| F-DL   | 3946      | 1/1   | 0.106     | 2.130     | 1.118          | 6.52  | 6.52       | 6.54  | 250E    | 0.112         |
| F671   | 3390      | 1/1   | 0.054     | 1.601     | 0.827          | 6.27  | 6.27       | 6.08  | 250E    | 0.161         |
| F671-1 | 3390      | 3/3   | 0.054     | 1.601     | 1.214          | 6.44  | 6.00       | 5.85  | 300E    | 0.218         |
| F692   | 3390      | 1/2   | 0.054     | 1.601     | 0.569          | 6.11  | 6.39       | 6.13  | 200E    | 0.094         |
| F692-R | 3151      | 2/3   | 0.057     | 1.829     | 0.943          | 6.27  | 6.27       | 6.14  | 200E    | 0.112         |
| F675   | 3151      | 1/3   | 0.057     | 1.829     | 0.500          | 6.00  | 6.44       | 6.19  | 200E    | 0.112         |
| F671-2 | 2655      | 3/3   | 0.069     | 2.572     | 1.946          | 6.45  | 6.01       | 5.87  | 300E    | 0.428         |
| F684   | 1857      | 2/3   | 0.113     | 5.778     | 2.946          | 6.35  | 6.35       | 6.15  | 200E    | 0.465         |
| F611   | 1857      | 1/3   | 0.113     | 5.778     | 1.530          | 6.07  | 6.52       | 6.19  | 125E    | 0.120         |
| F652   | 1874      | 1/2   | 0.112     | 5.660     | 1.961          | 6.18  | 6.47       | 6.17  | 150E    | 0.207         |

**Table 25:** Fuse coefficients against Table III of the paper. “ $b_i$  rev.”: index  $i$  counted from the source, which reproduces the paper’s values in R1’s zone.

| Node | Fault | $I_{R1}$ | R1 fast (s)   | R1 delayed (s) | $I_{R2}$ | Dir. | R2 fast (s)   | R2 delayed (s) |
|------|-------|----------|---------------|----------------|----------|------|---------------|----------------|
|      |       | (A)      | model / paper | model / paper  | (A)      |      | model / paper | model / paper  |
| 632  | LLL   | 4734     | 0.081 / 0.070 | 1.627 / 1.398  | 1813     | rev  | 0.051 / 0.045 | 1.416 / 0.514  |
| 633  | LLL   | 3970     | 0.106 / 0.094 | 2.111 / 1.880  | 1500     | rev  | 0.060 / 0.052 | 2.001 / 0.706  |
| 645  | LL    | 3124     | 0.155 / 0.105 | 3.095 / 2.095  | 1232     | rev  | 0.075 / 0.087 | 3.002 / 1.659  |
| 646  | LL    | 2785     | 0.189 / 0.128 | 3.770 / 2.555  | 1114     | rev  | 0.086 / 0.103 | 3.771 / 2.046  |
| DL   | LLL   | 3954     | 0.106 / 0.087 | 2.123 / 1.733  | 1972     | rev  | 0.048 / 0.076 | 1.213 / 1.402  |
| 671  | LLL   | 3412     | 0.134 / 0.105 | 2.677 / 2.091  | 3391     | fwd  | 0.054 / 0.068 | 1.600 / 1.113  |
| 692  | LLL   | 3412     | 0.134 / 0.115 | 2.677 / 2.307  | 3391     | fwd  | 0.054 / 0.072 | 1.600 / 1.258  |
| 675  | LLL   | 3090     | 0.158 / 0.121 | 3.152 / 2.414  | 3050     | fwd  | 0.059 / 0.074 | 1.941 / 1.332  |
| 684  | LL    | 2543     | 0.222 / 0.125 | 4.437 / 2.507  | 2543     | fwd  | 0.072 / 0.075 | 2.813 / 1.394  |
| 680  | LLL   | 2789     | 0.188 / 0.137 | 3.761 / 2.747  | 2757     | fwd  | 0.066 / 0.079 | 2.381 / 1.555  |
| 611  | LG    | 1814     | 0.435 / 0.326 | 8.709 / 6.523  | 1718     | fwd  | 0.128 / 0.146 | 6.986 / 4.385  |
| 652  | LG    | 1795     | 0.446 / 0.676 | 8.919 / 13.525 | 1761     | fwd  | 0.123 / 0.143 | 6.570 / 9.370  |

**Table 26:** Recloser operating times against Table IV of the paper.

| Node | Fault | Fuse   | Size | $I_{fuse}$ | $t_{MMT}$ (s) |       | $t_{TCT}$ | Last fast | Margin | Status |
|------|-------|--------|------|------------|---------------|-------|-----------|-----------|--------|--------|
|      |       |        |      | (A)        | model         | paper | (s)       | trip (s)  | (s)    |        |
| 633  | LLL   | F633   | 400E | 5426       | 0.149         | 0.163 | 0.235     | 0.106     | +0.043 | held   |
| 645  | LL    | F632   | 500E | 4191       | 0.613         | 0.276 | 0.970     | 0.155     | +0.458 | held   |
| 646  | LL    | F646   | 400E | 3672       | 0.415         | 0.181 | 0.684     | 0.189     | +0.226 | held   |
| DL   | LLL   | F-DL   | 400E | 5902       | 0.121         | 0.248 | 0.193     | 0.106     | +0.015 | held   |
| 692  | LLL   | F671-1 | 400E | 3391       | 0.525         | 0.175 | 0.891     | 0.054     | +0.471 | held   |
| 675  | LLL   | F692-R | 250E | 5022       | 0.065         | 0.185 | 0.106     | 0.059     | +0.006 | held   |
| 684  | LL    | F671-2 | 300E | 3963       | 0.148         | 0.205 | 0.230     | 0.072     | +0.076 | held   |
| 611  | LG    | F684   | 200E | 2236       | 0.271         | 0.417 | 0.414     | 0.128     | +0.143 | held   |
| 652  | LG    | F652   | 150E | 2239       | 0.138         | 0.233 | 0.193     | 0.123     | +0.015 | held   |

**Table 27:** Fuse times against Table IV of the paper.

| Fault       | 632 | 633 | 645 | 646 | DL | 671 | 692 | 675 | 680 | 684 | 611 | 652 |
|-------------|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|
| LG (model)  | ✓   | ×   | ×   | ×   | ×  | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   |
| LG (paper)  | ✓   | ✓   | ×   | ✓   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   | ✓   |
| LL (model)  | ✓   | ×   | ✓   | ×   | ×  | ✓   | ✓   | ×   | ✓   | ✓   | —   | —   |
| LL (paper)  | ✓   | ✓   | ×   | ×   | ✓  | ✓   | ✓   | ✓   | ✓   | ✓   | —   | —   |
| LLG (model) | ✓   | ×   | ✓   | ×   | ×  | ✓   | ✓   | ×   | ✓   | ✓   | —   | —   |
| LLG (paper) | ✓   | ×   | ×   | ×   | ✓  | ✓   | ✓   | ×   | ✓   | ✓   | —   | —   |
| LLL (model) | ✓   | ×   | —   | —   | ×  | ✓   | ✓   | ×   | ✓   | —   | —   | —   |
| LLL (paper) | ✓   | ×   | —   | —   | ✓  | ✓   | ✓   | ×   | ✓   | —   | —   | —   |

**Table 28:** Coordination status with DG and a conventional R2: model against the paper's Fig. 14.

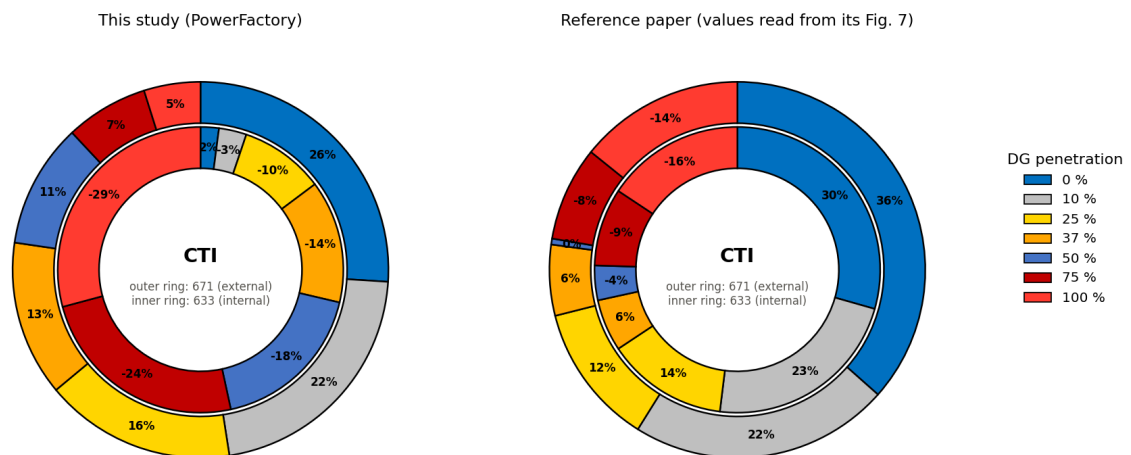
| Event (time in s)              | Paper (read off Fig. 10) | Model, DG out | Model, DG in |
|--------------------------------|--------------------------|---------------|--------------|
| Fault initiated                | 0.30                     | 0.30          | 0.30         |
| R2 fast trip 1                 | $\approx 0.46$           | 0.420         | 0.426        |
| R2 reclose 1                   | $\approx 0.65$           | 0.620         | 0.626        |
| R2 fast trip 2                 | $\approx 0.80$           | 0.740         | 0.753        |
| R2 reclose 2                   | $\approx 1.00$           | 0.940         | 0.953        |
| Fuse F671-2 melts              | –                        | 1.248         | 0.752        |
| Fuse operation / clears        | 1.21                     | 1.613         | 1.253        |
| Fuse heat after the fast shots | –                        | 45 %          | 105 %        |

**Table 29:** Reclosing sequence of the EMT simulation against the paper’s Fig. 10.

## DG penetration (Fig. 7 of the paper)

The paper’s Fig. 7 gives the CTI at nodes 633 and 671 as shares of a doughnut chart. The trend is the same in both: the CTI falls monotonically as the penetration rises. At 671 the shares of the first four levels are close to the paper’s. The levels at which the CTI changes sign differ: in the paper both nodes cross zero at about 40–50 %; in this study 633 crosses at about 4 % and 671 does not cross up to 100 %. The crossing depends on the starting margin, which depends on fuse sizes and settings that the paper does not publish.

Fig. 7 - percentage variation of the CTI with DG penetration (each ring = 100 %; negative = fuse melts before the recloser’s fast trip)



**Figure 23:** CTI against DG penetration: this study (left) and the values read from the paper’s Fig. 7 (right).



| DG (%) | Node 633 (ring %) |       | Node 671 (ring %) |       |
|--------|-------------------|-------|-------------------|-------|
|        | this study        | paper | this study        | paper |
| 0      | +2.1              | +30   | +26.0             | +36   |
| 10     | −3.1              | +23   | +21.5             | +22   |
| 25     | −9.6              | +14   | +16.4             | +12   |
| 37     | −13.9             | +6    | +13.4             | +6    |
| 50     | −17.9             | −4    | +10.7             | +0    |
| 75     | −24.2             | −9    | +7.1              | −8    |
| 100    | −29.2             | −16   | +4.9              | −14   |

**Table 30:** CTI shares against DG penetration: this study and the paper’s Fig. 7.